

# Quintile Return Book

Average monthly return of each factor's quintile portfolios and the long / short quintiles selected from them. 270 factors, ordered by in-sample L–S CAGR (last 48 months, net of cost).

## HOW TO READ

Long Short Neutral

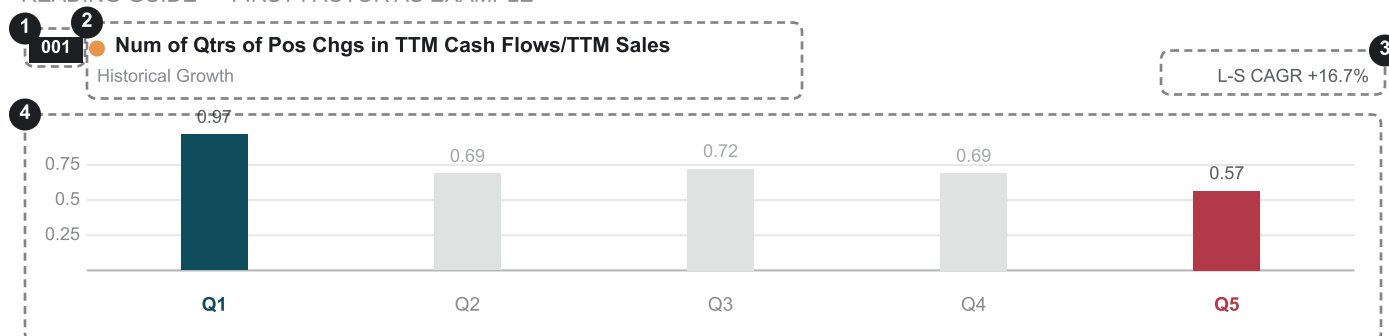
Quintiles are selected as long or short by performance against a spread-based threshold — not fixed to Q1 / Q5. Grey quintiles are left out of the portfolio.

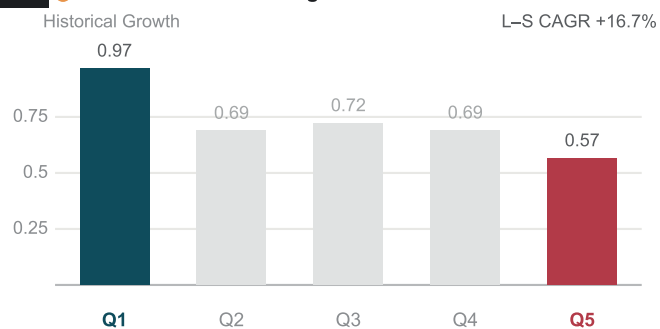
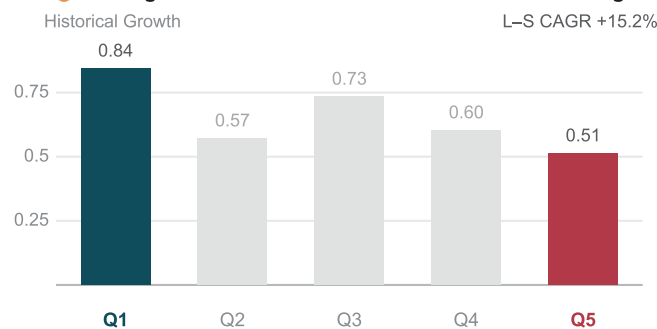
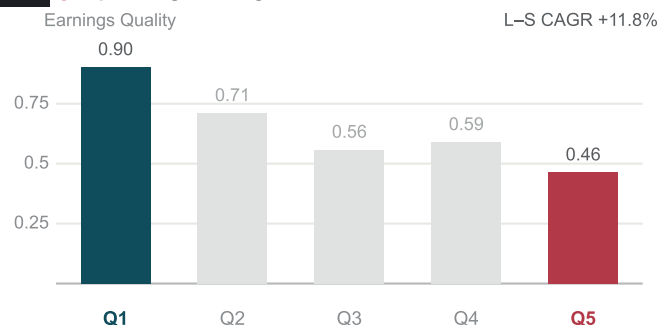
## FACTOR STYLES

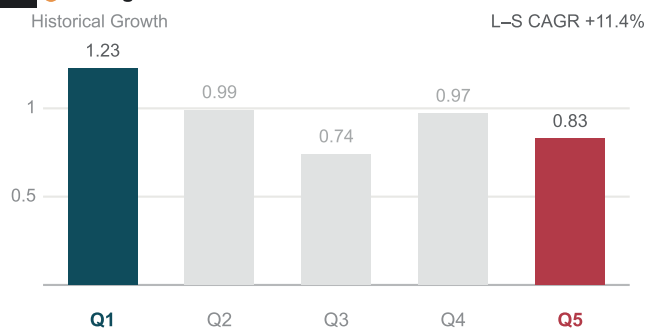
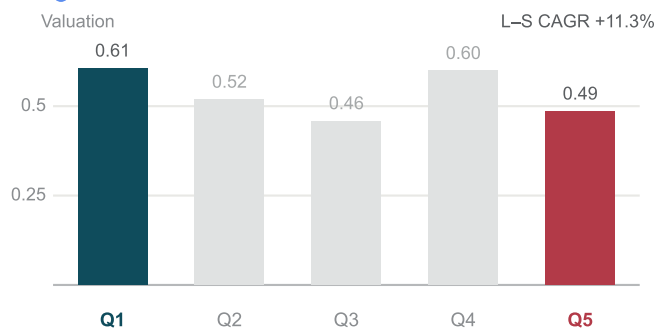
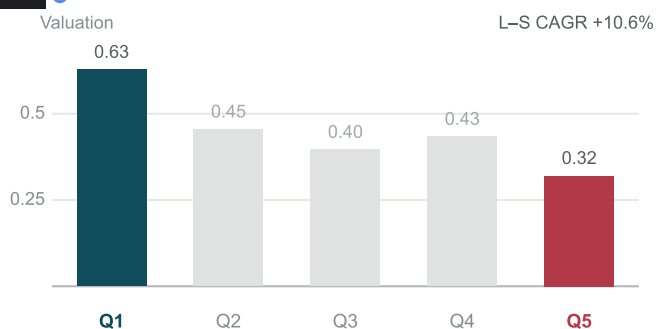
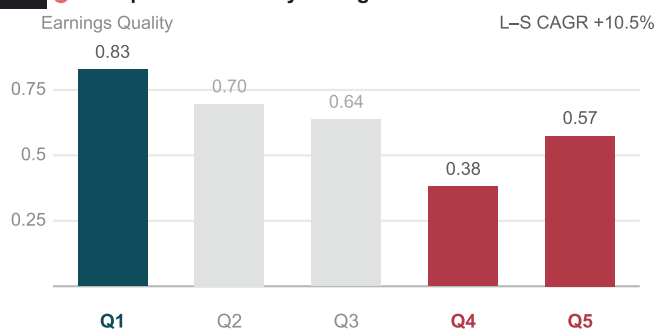
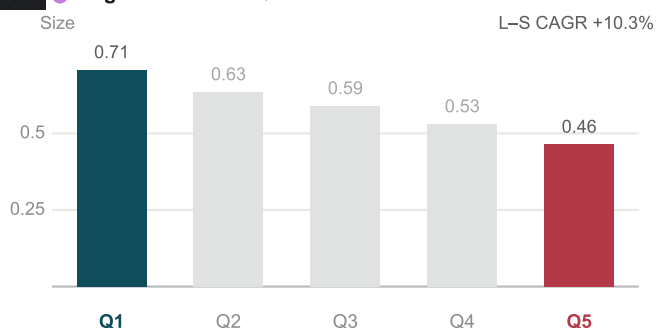
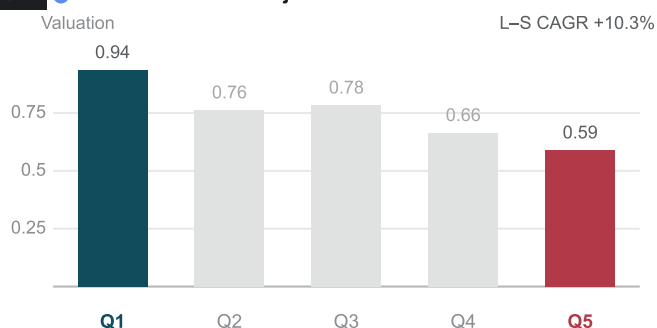
|                      |    |                   |    |
|----------------------|----|-------------------|----|
| Valuation            | 41 | Price Momentum    | 56 |
| Earnings Quality     | 55 | Size              | 1  |
| Analyst Expectations | 24 | Historical Growth | 55 |
| Capital Efficiency   | 28 | Volatility        | 10 |

**001** In current model portfolio (50 factors)

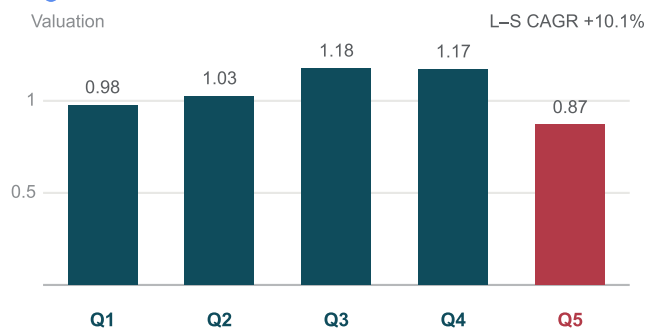
## READING GUIDE — FIRST FACTOR AS EXAMPLE



**001 ● Num of Qtrs of Pos Chgs in TTM Cash Flows/TTM Sales****002 ● Trailing 12-Month Receivables and Inventories/Trailing 12-M...****003 ● Pretax Income Per Share to Price****004 ● Current Liability to Price****005 ● Number of Consecutive Qtrs of Pos. Changes in TTM Cash Flo****006 ● Operating Earnings to Assets Ratio**

**007 ● Change of Asset Turnover****008 ● Assets to Price Ratio****009 ● Sales to Price Ratio****010 ● Unexpected Inventory Change****011 ● Log of Total Last Quarter Assets****012 ● Inverse P/E Ratio Adj for Growth and Yield**

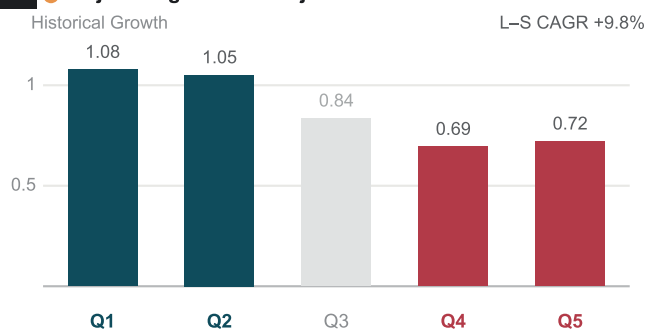
## 013 ● Indicated Annual Dividends to Price Ratio



## 014 ● Cash to Price



## 015 ● Adj 1Y Chg in Asset Adjusted FCF



## 016 ● Long Term Debt to Equity Ratio

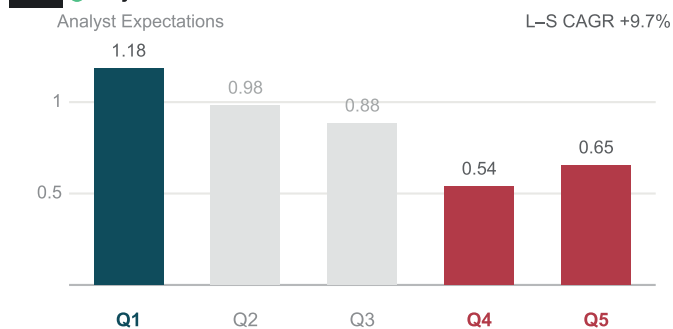
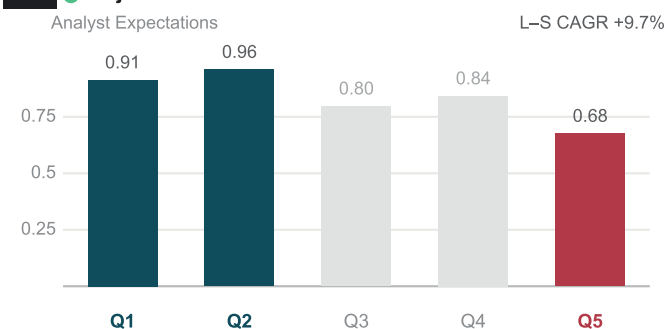
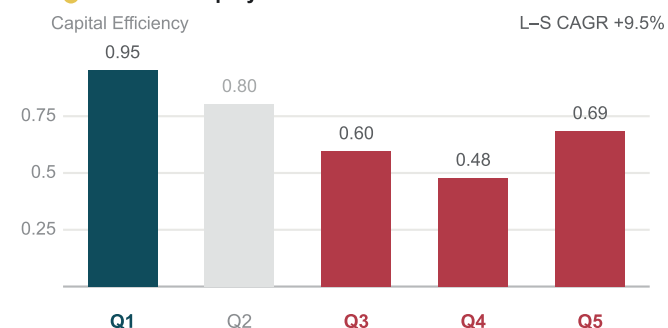
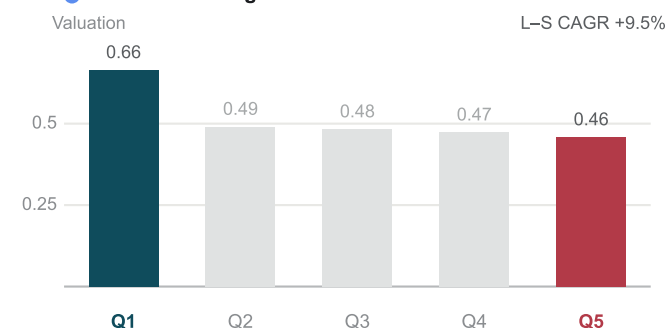


## 017 ● Return on Assets

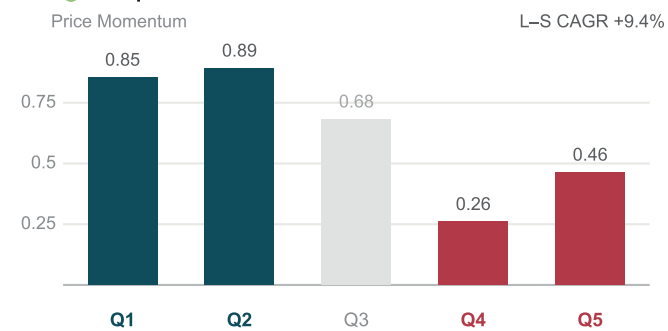


## 018 ● Number of Consecutive Qtrs of Positive Changes in TTM Quart...



**019 ● Gross Profit Margin****020 ● Adjusted 3M Revision in FY1 EPS Estimate****021 ● Adjusted Inverse PEG****022 ● Analyst Dispersion for FY2 EPS****023 ● Return on Equity****024 ● Forward Earnings to Price**

## 025 ● Sharpe Ratio



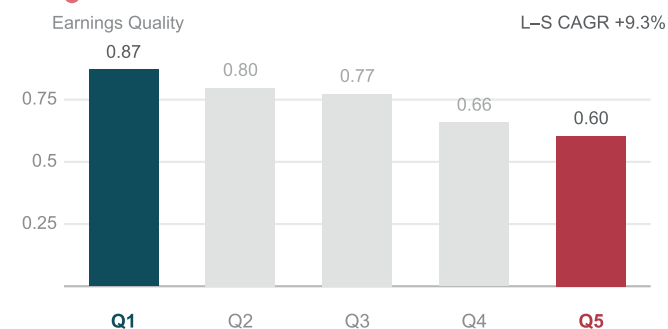
## 026 ● 12M Realized Price Volatility



## 027 ● Return on Invested Capital



## 028 ● Quick Ratio

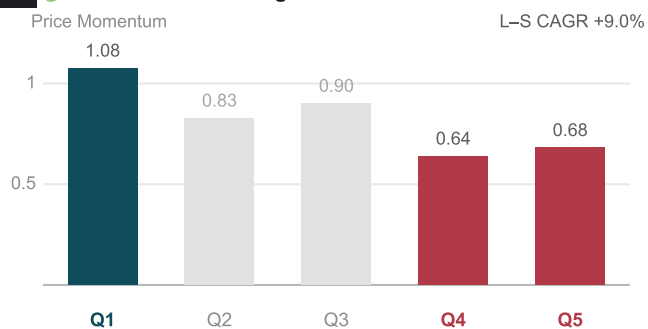
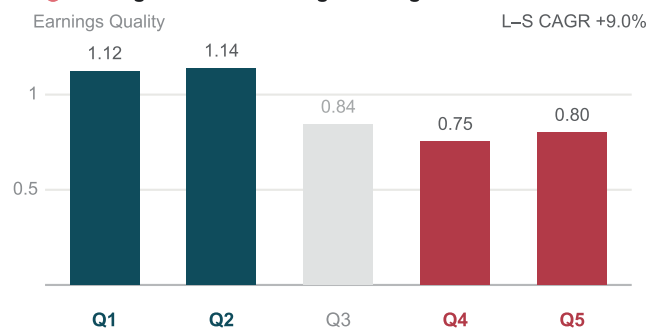
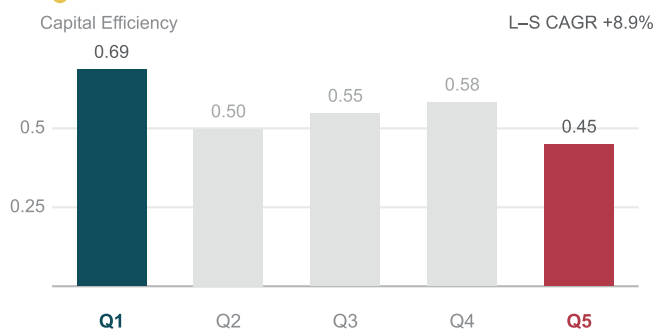
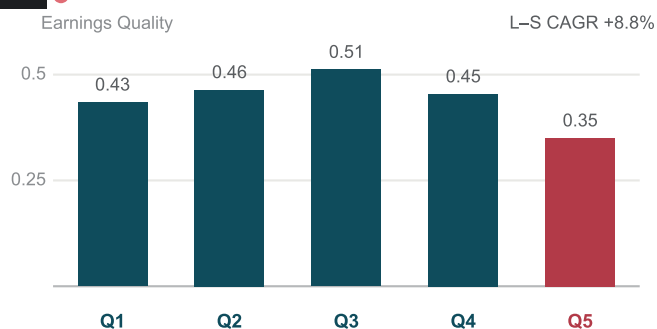
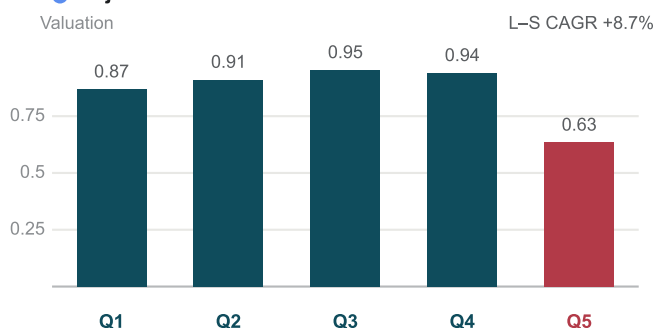
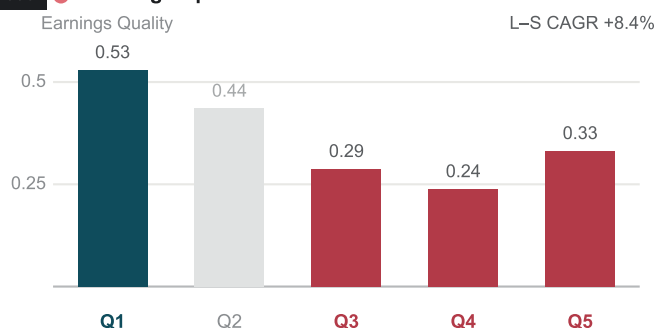


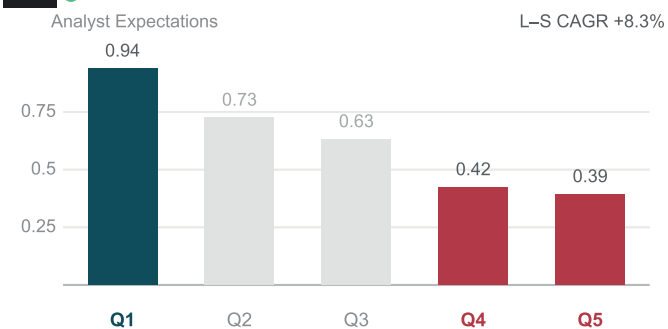
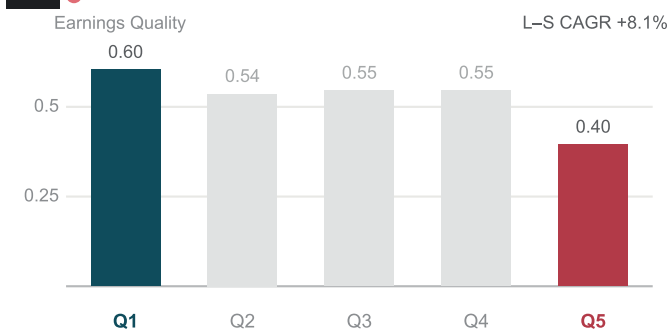
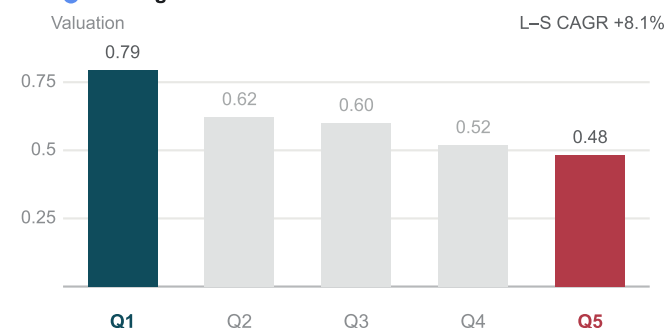
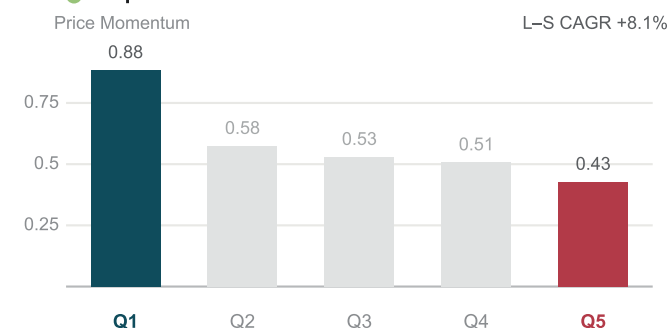
## 029 ● 3Y Chg in Operating Cash Flow to Price



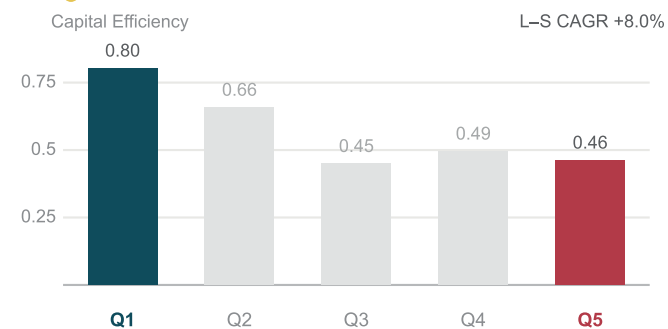
## 030 ● Operating Earnings to Price Ratio



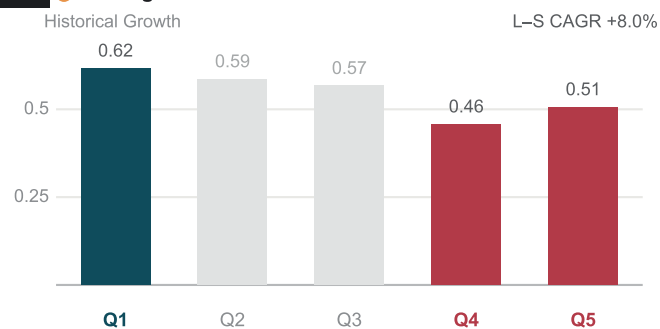
**031 ● Short Interest Trading Volume****032 ● Trailing 12-Month Earnings/Trailing 12-Month Sales****033 ● Tobin's Q Ratio****034 ● ROA 20 Qtr Standard Deviation****035 ● Adj Growth Flow to Price****036 ● Working Capital to Sales**

**037 ● 3M Revision in FY1 EPS Estimate****038 ● Adjusted Forward Earnings to Price****039 ● Normalized EP****040 ● Balance Sheet Accruals****041 ● Earnings to Price****042 ● Slope of 52W Price Trend**

## 043 ● EBITDA to Assets



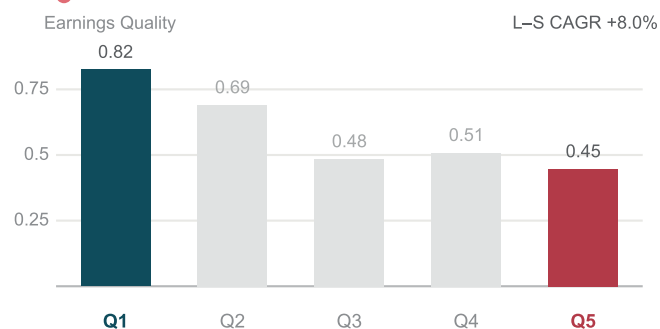
## 044 ● 3Y Chg in Cash Flow to Price



## 045 ● Consecutive Quarters of Declines in TTM Overhead/Sales



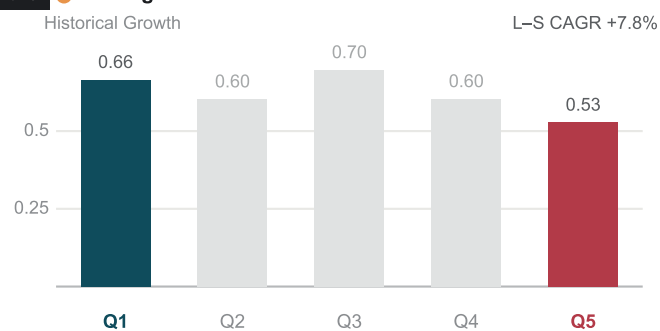
## 046 ● Cash Flow to Total Asset



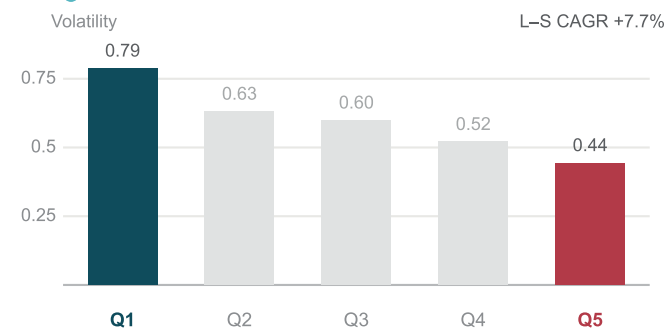
## 047 ● Book Leverage



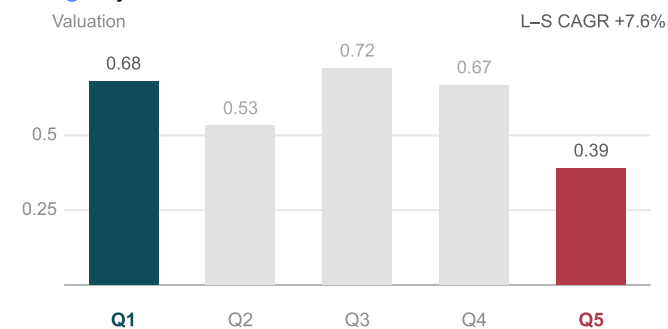
## 048 ● 1Y Chg in Sales to Price



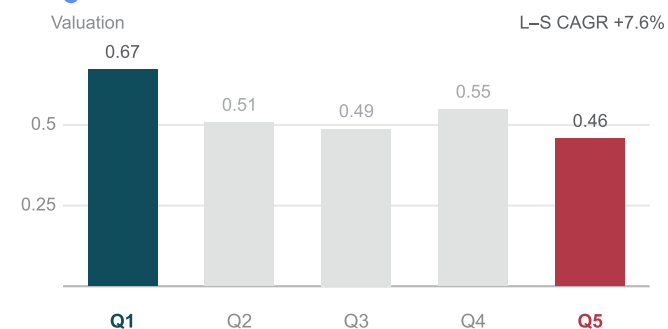
## 049 ● 60M CAPM Beta



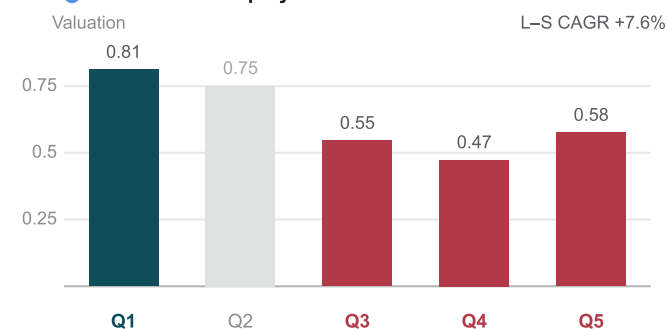
## 050 ● Adj Free Cash Flow to Price



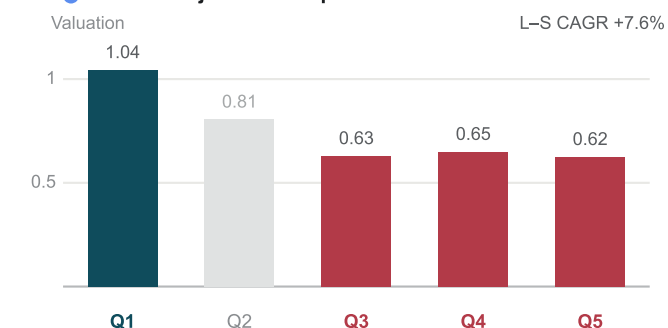
## 051 ● Sales to EV Ratio



## 052 ● Cash Flow to Equity

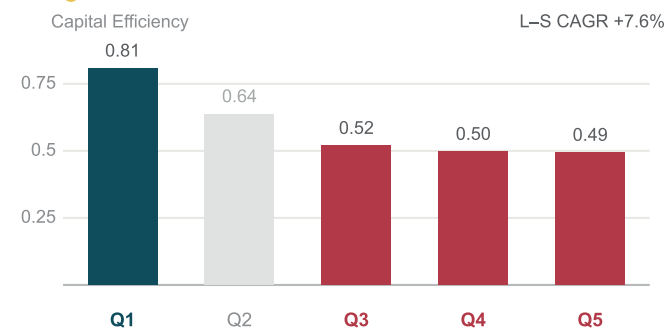
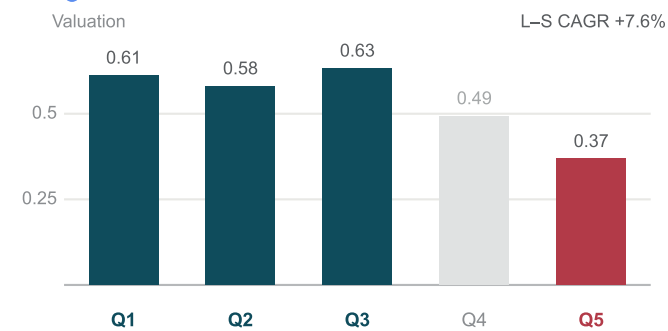
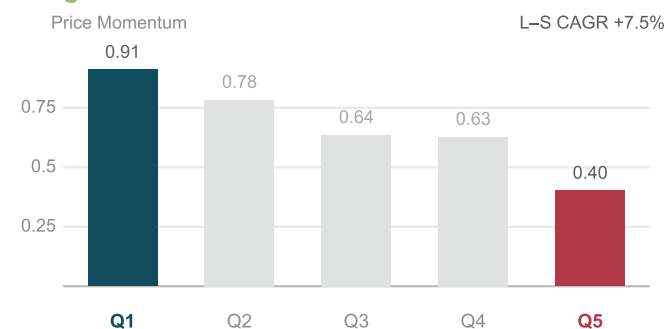
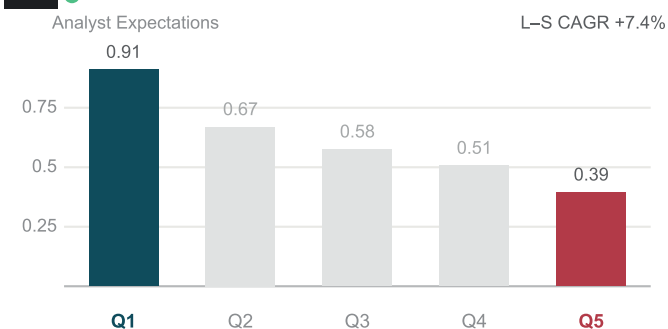
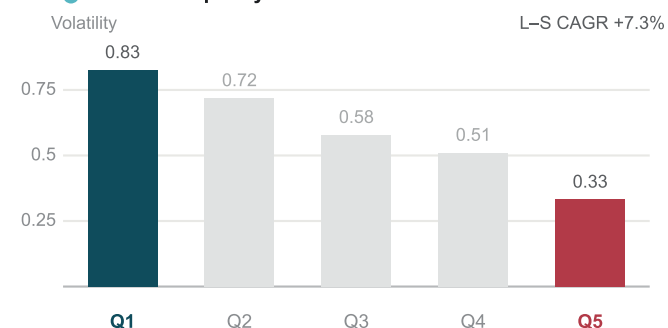
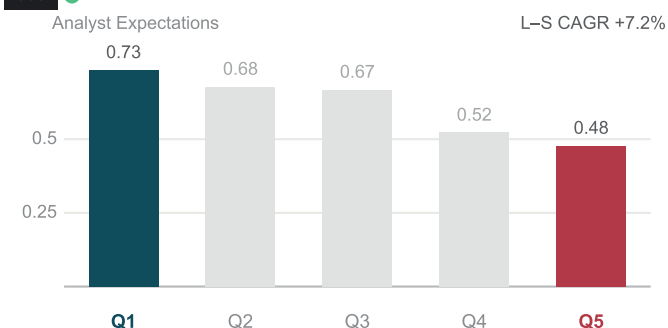


## 053 ● Cash to Adjusted Enterprise Value

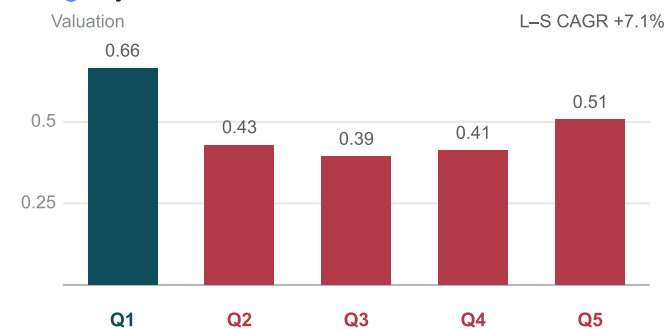


## 054 ● Expected LTG

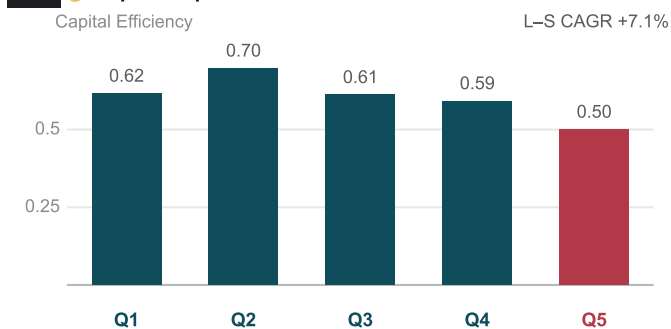


055 ● **EBIT to Assets**056 ● **Dividends to Price Ratio**057 ● **Short Interest Ratio**058 ● **3M Revision in FY2 EPS Estimate**059 ● **Amihud Illiquidity Measure**060 ● **Inverse PEG**

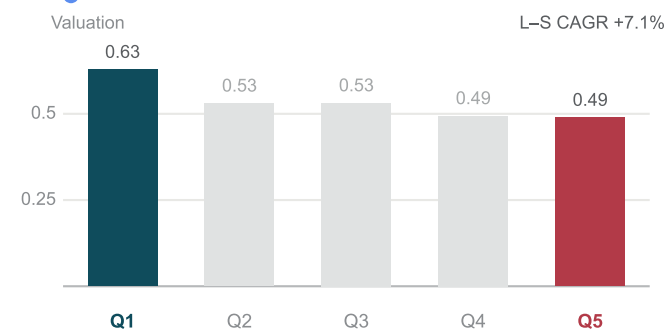
## 061 ● Adjusted EBIT to Price



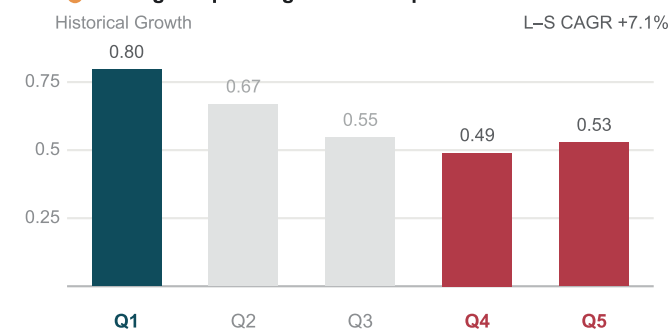
## 062 ● Capital Expenditure to Total Assets



## 063 ● Net Cash Flow to Price



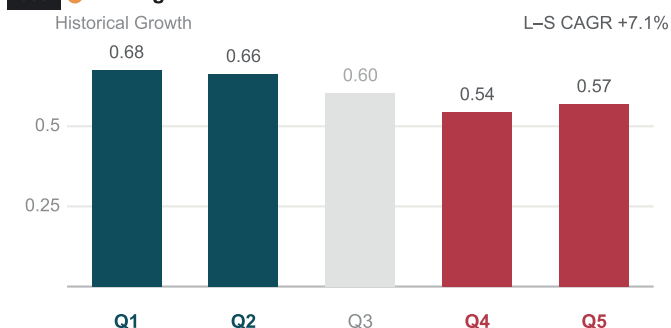
## 064 ● 3Y Chg in Operating Cash Flow per Share



## 065 ● 1Y Chg in Operating Margin



## 066 ● 1Y Chg in Cash Flow to Price



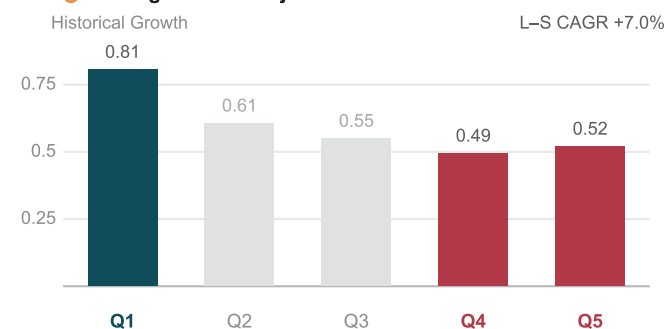
## 067 ● 50 Day to 200 Day Stock Price



## 068 ● Adjusted Book to Enterprise Value



## 069 ● 1Y Chg in Asset Adjusted EPS



## 070 ● Earnings Surprise

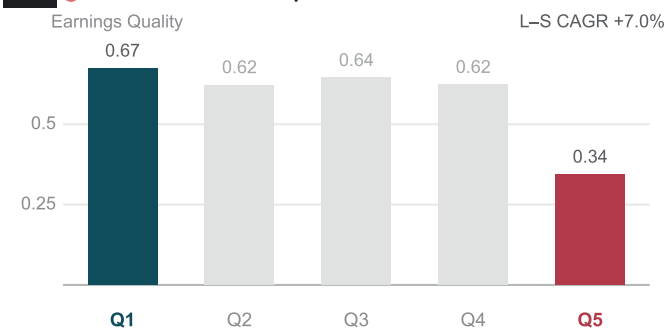
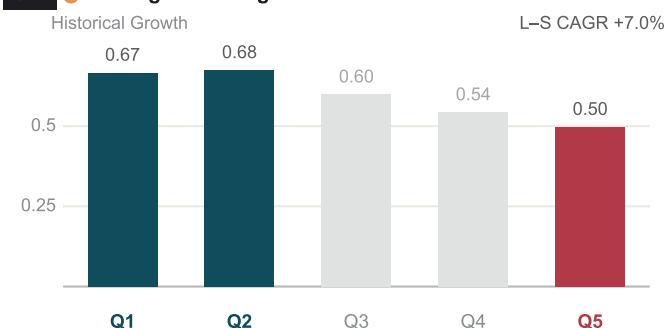


## 071 ● Street Revision Magnitude



## 072 ● Cash Flow to Invested Capital

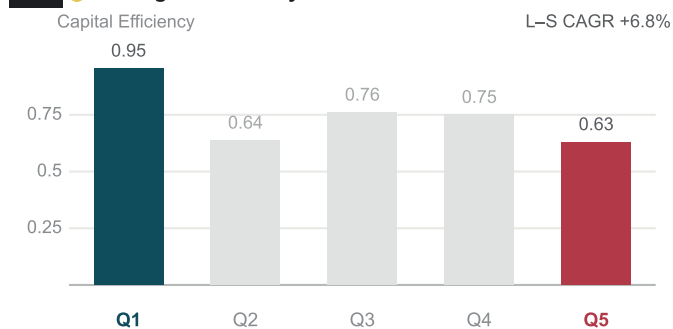


**073 ● Sales to Invested Capital****074 ● 39W Lagged Return****075 ● Operating Cash Flow to Invested Capital****076 ● Adjusted 3Y Chg in Asset Adjusted Free Cash Flow****077 ● 1Y Chg in Earnings to Price****078 ● Operating Cash Flow to Asset**

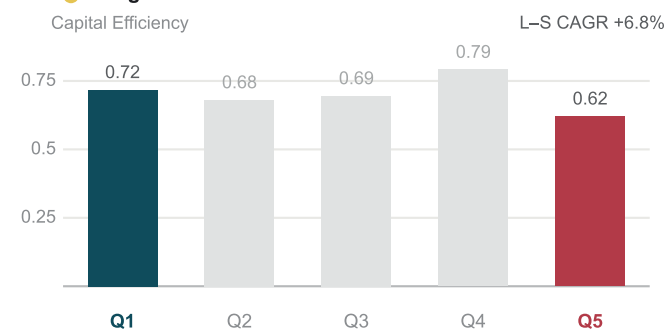
## 079 ● EBITDA to Enterprise Value



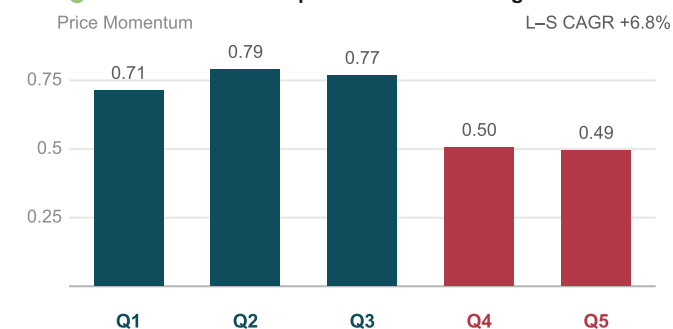
## 080 ● 1Y Chg in Stock Buybacks



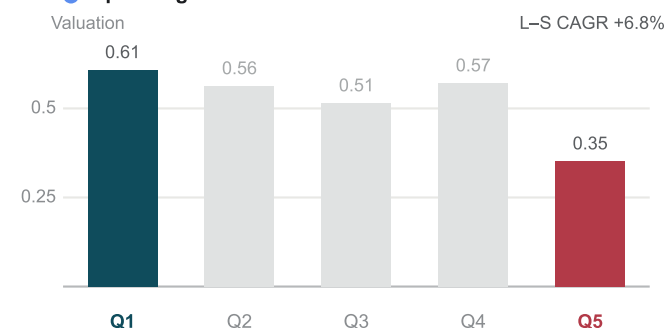
## 081 ● Long Term Debt to Assets Ratio



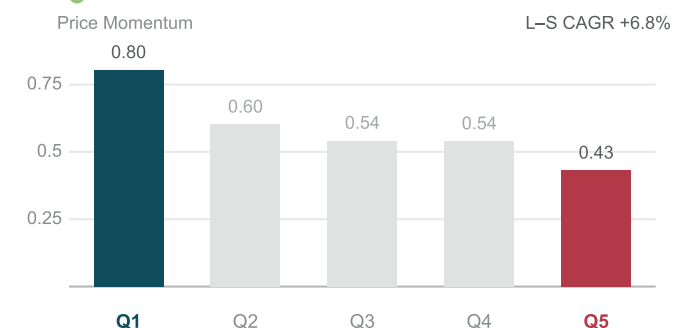
## 082 ● Institution Ownership Level - Active Manager



## 083 ● Operating Cash Flow to Price



## 084 ● 15/36 Week Price Ratio



## 085 ● 26W Relative Price Strength



## 086 ● Adj 1Y Chg in Earnings per Share



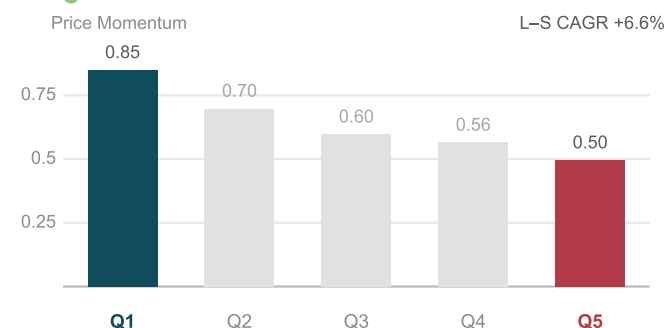
## 087 ● 90 Day Coefficient of Variation



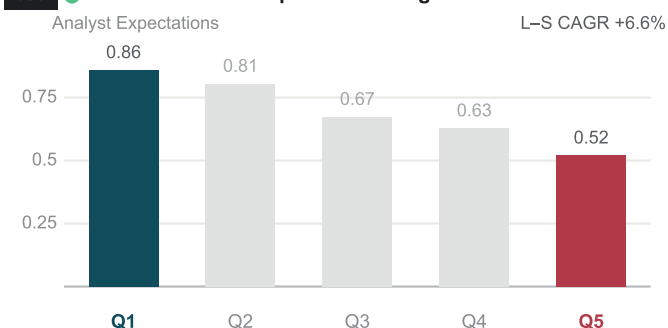
## 088 ● Solvency Ratio

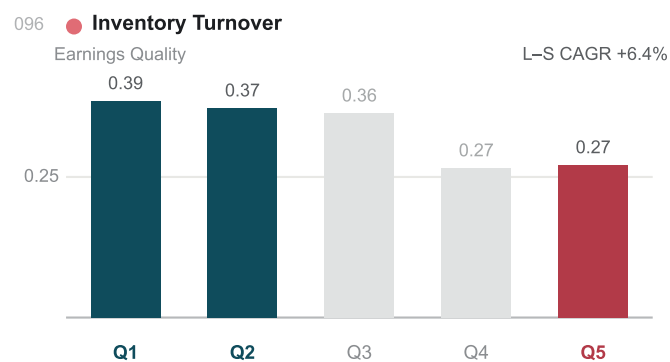
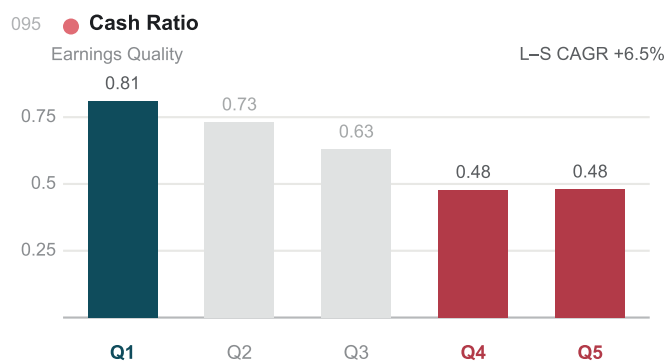
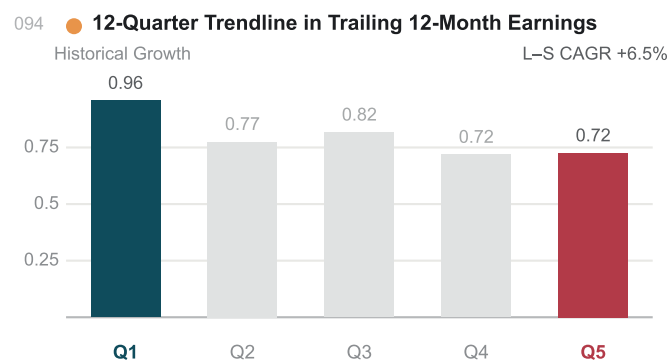
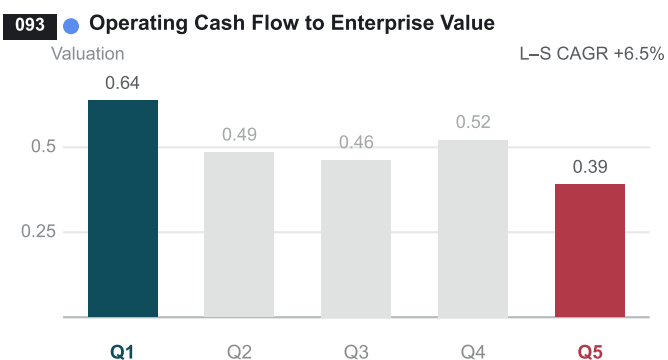
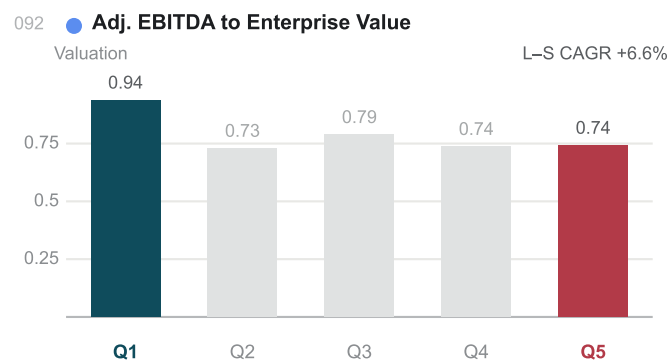
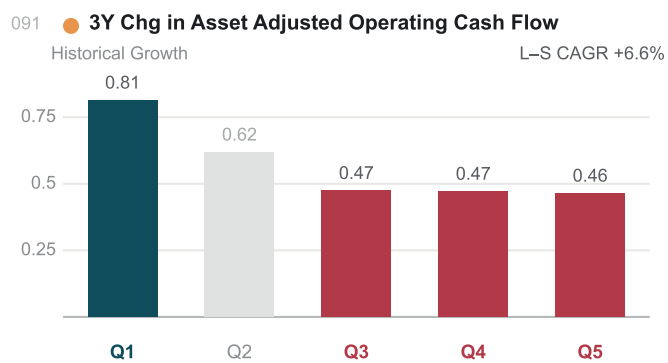


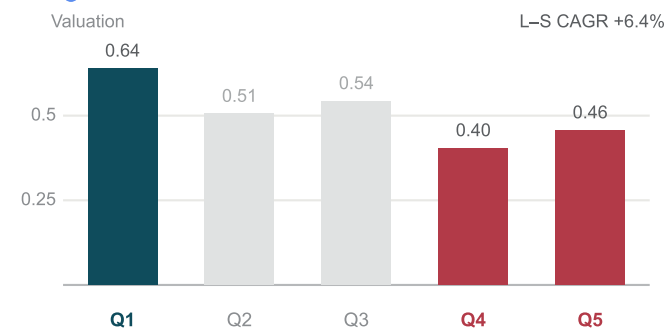
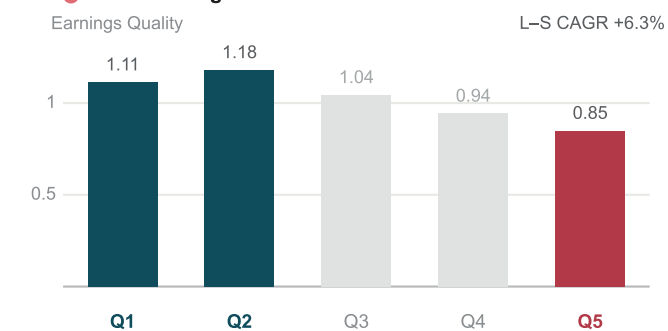
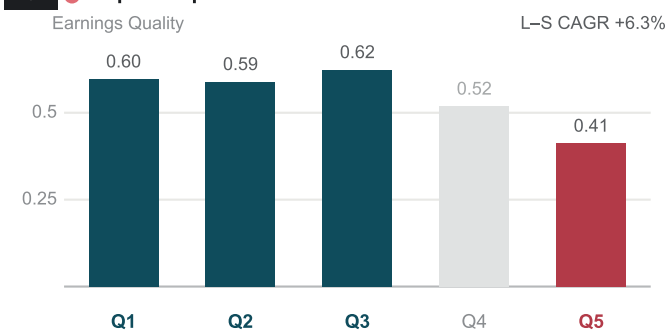
## 089 ● 9 Month Price Momentum



## 090 ● Standardized Unexpected Earnings





097 ● **Book to Price**098 ● **6M Chg in Target Price**099 ● **Adjusted Revision Magnitude**100 ● **Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1**101 ● **EBITDA Margin**102 ● **Capital Expenditure to Sales**

## 103 ● Free Cash Flow to Enterprise Value



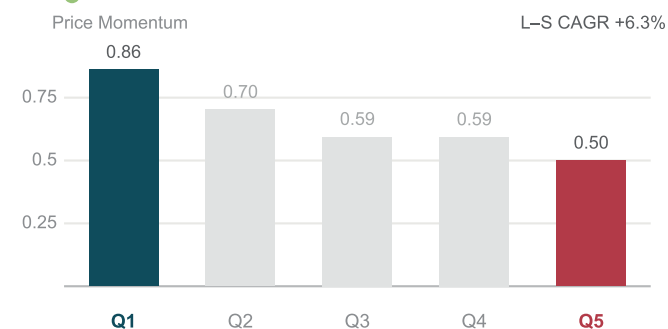
## 104 ● 1M Realized Price Volatility



## 105 ● EBITDA to Price



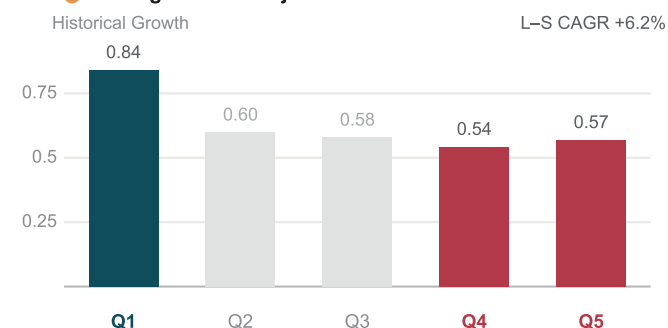
## 106 ● 4W to 52W Price Oscillator



## 107 ● Percent Above 260-Day Low



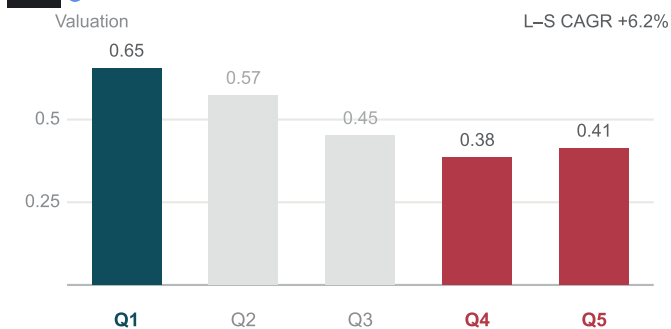
## 108 ● 1Y Chg in Asset Adjusted Cash Flow



## 109 ● Growth Flow to Price



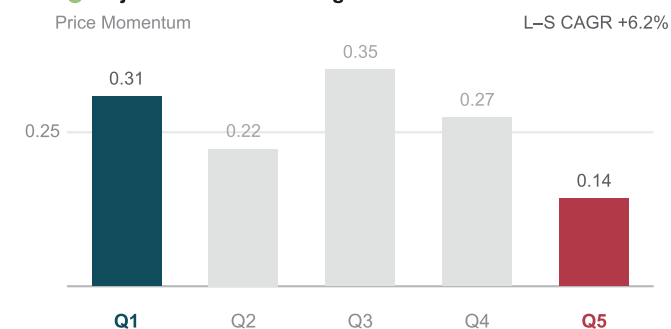
## 110 ● Cash to EV Ratio



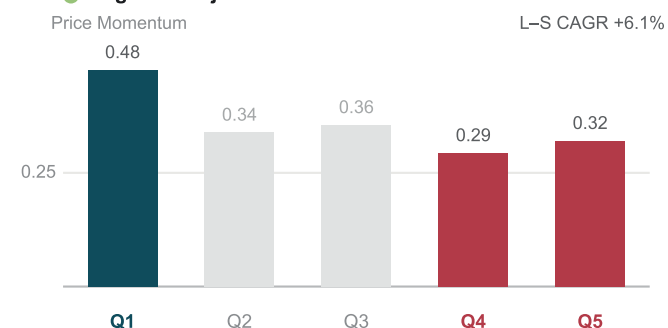
## 111 ● Year over Year Change in SGA to Sales



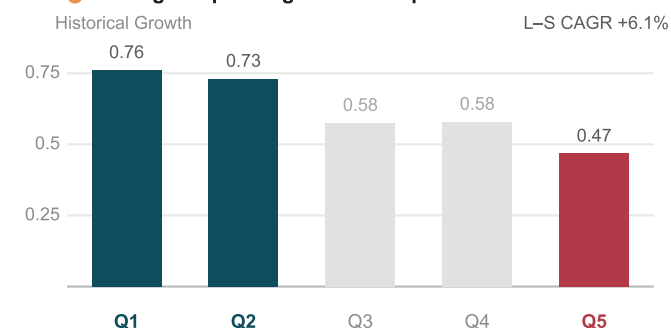
## 112 ● Adjusted 6 Month Average Share Turn Over



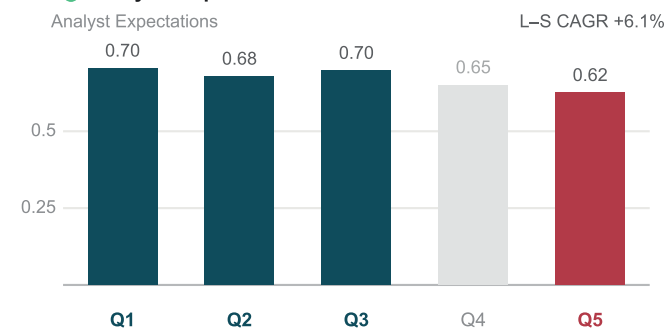
## 113 ● Log of Unadjusted Stock Price



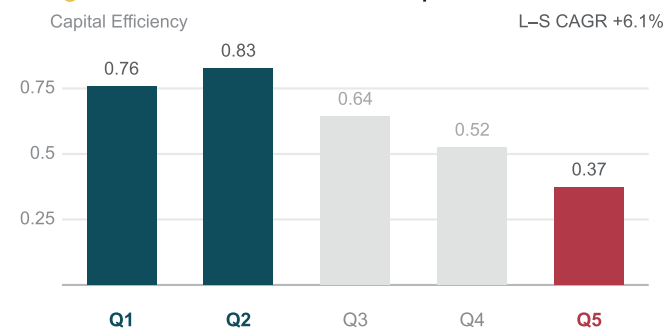
## 114 ● 1Y Chg in Operating Cash Flow per Share



## 115 ● Analyst Dispersion for FY1 EPS



## 116 ● Cash Flow Return on Invested Capital



## 117 ● 1Y Chg in Asset Adjusted Free Cash Flow



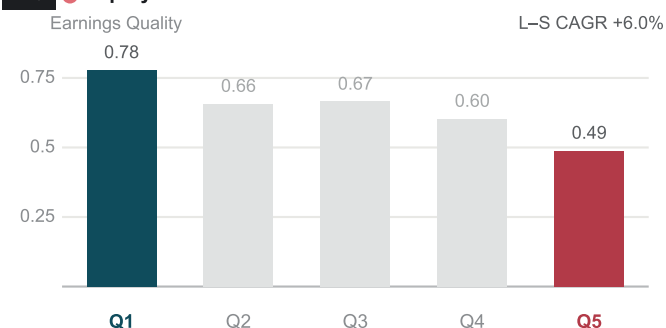
## 118 ● Stability of Change in Number of Holders

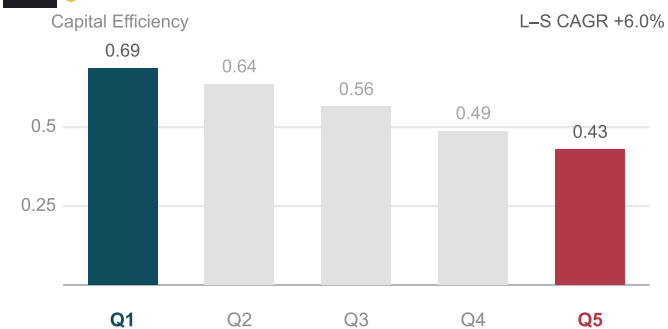
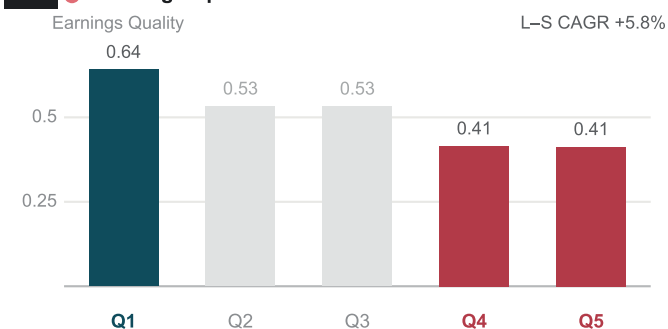
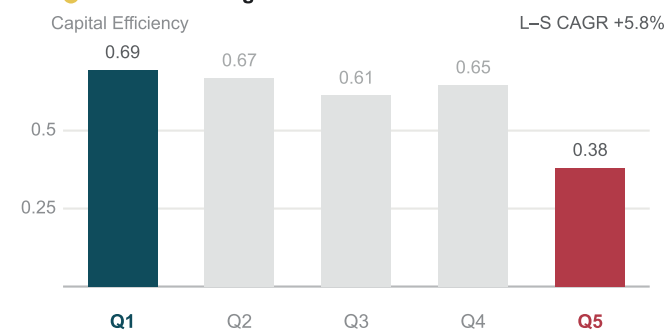


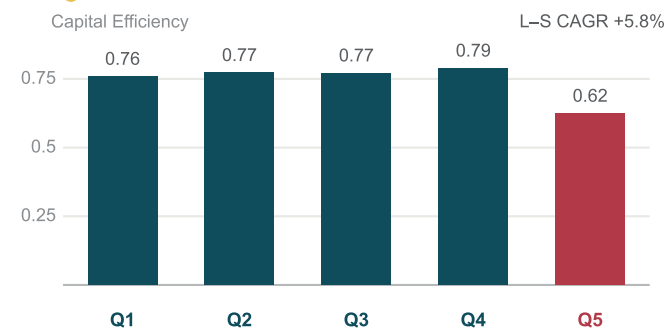
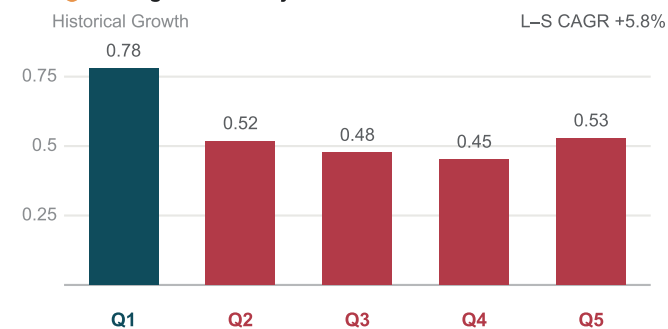
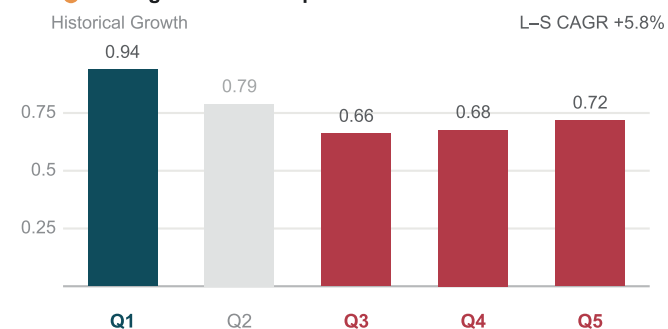
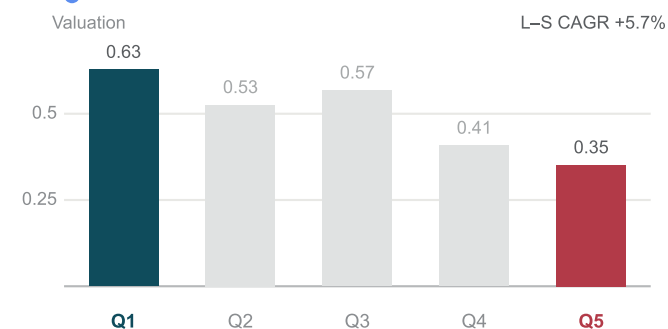
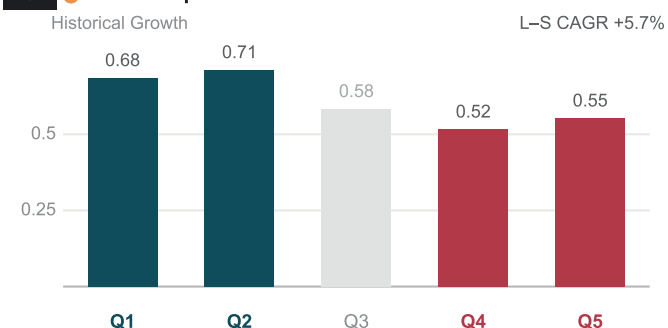
## 119 ● Operating Cash Flow Ratio

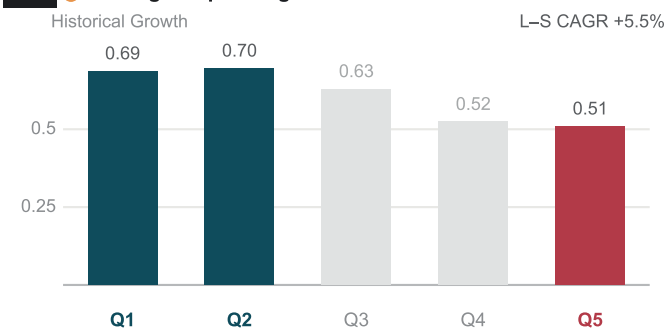
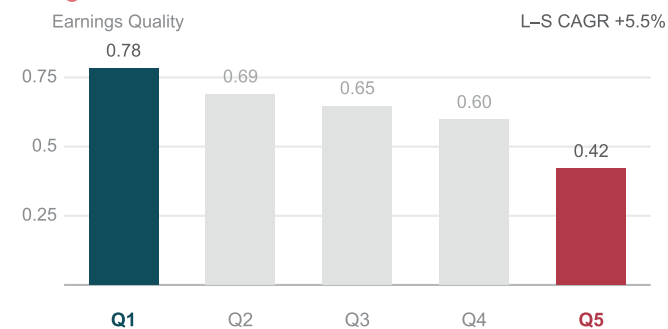


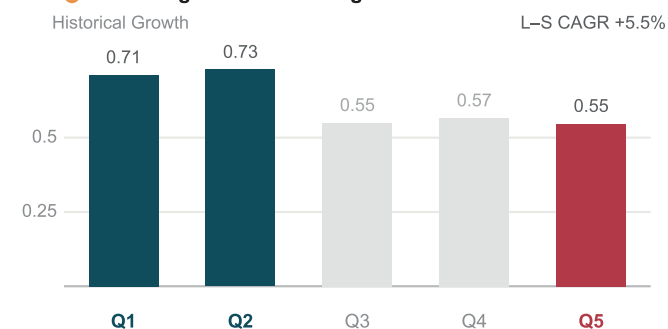
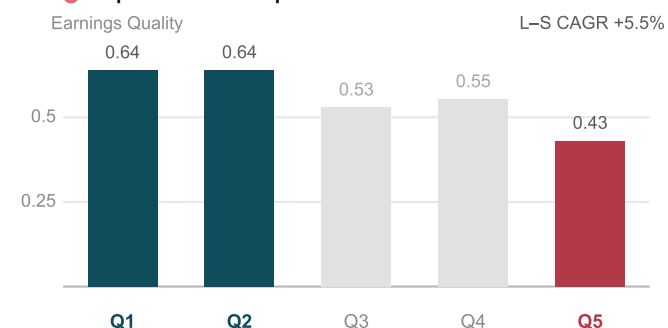
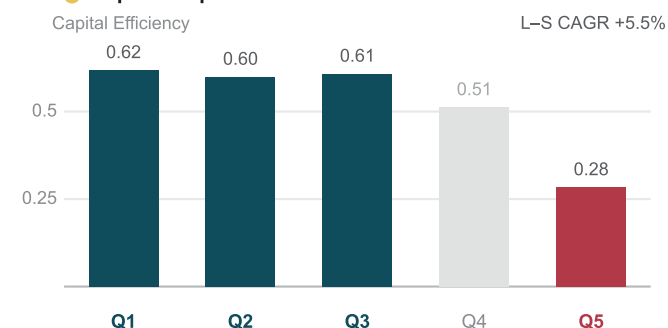
## 120 ● Equity Turnover



**121 ● Receivables Turnover Ratio****122 ● Adjusted Interest Coverage Ratio****123 ● 3Y Chg in Free Cash Flow to Price****124 ● Working Capital Accruals****125 ● Interest Coverage Ratio****126 ● Forward Free Cash Flow to Price Ratio**

127 ● **Debt to Assets Ratio**128 ● **3Y Chg in Asset Adjusted Cash Flow**129 ● **3Y Chg in Cash Flow per Share**130 ● **Free Cash Flow to Price**131 ● **Dividend per Share Growth**132 ● **12-Quarter Trendline in Trailing 12-Month Cash Flows**

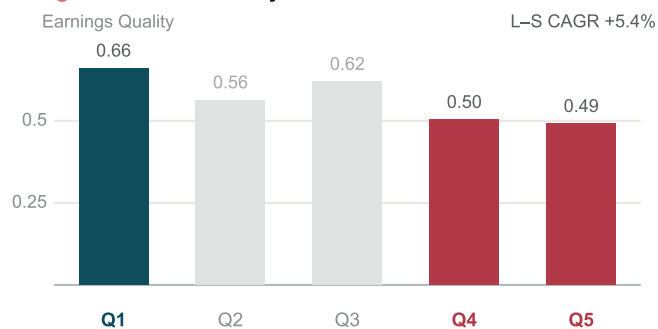
**133 ● 1Y Chg in Operating Cash Flow to Price**

**134 ● Cash Burn Rate**

**135 ● 52-Week Volume Price Trend with 20-Day Lag**

**136 ● 3Y Average Annual Earnings Growth**

**137 ● Depreciation to Capex Ratio**

**138 ● Capital Acquisition Ratio**


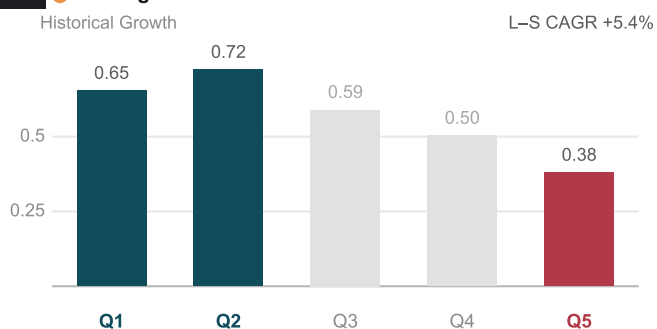
## 139 ● Free Cash Flow Stability



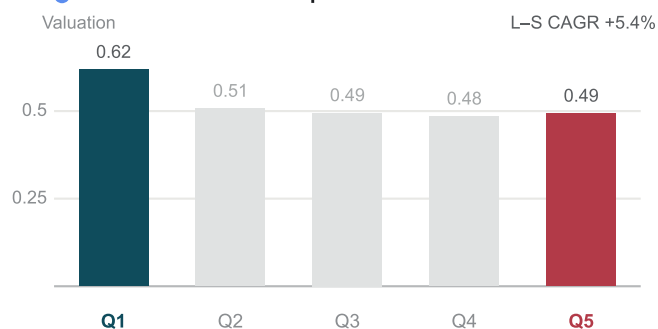
## 140 ● Net Income Stability



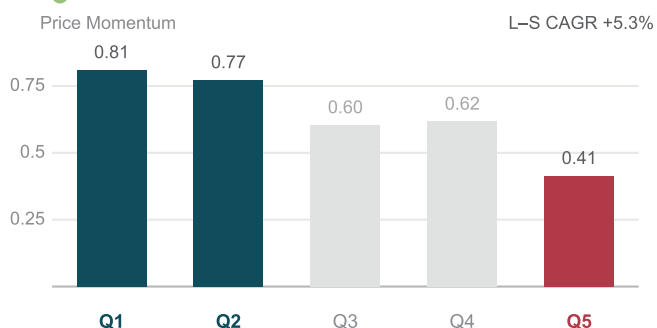
## 141 ● 1Y Chg in Free Cash Flow to Price



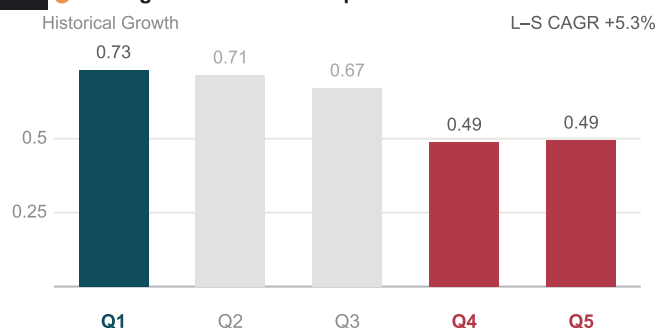
## 142 ● Net Cash Flow to Enterprise Value

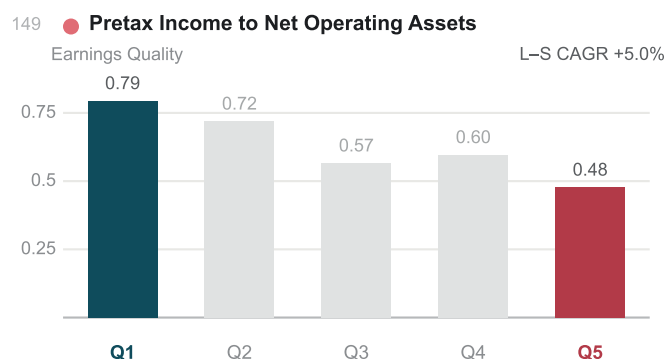
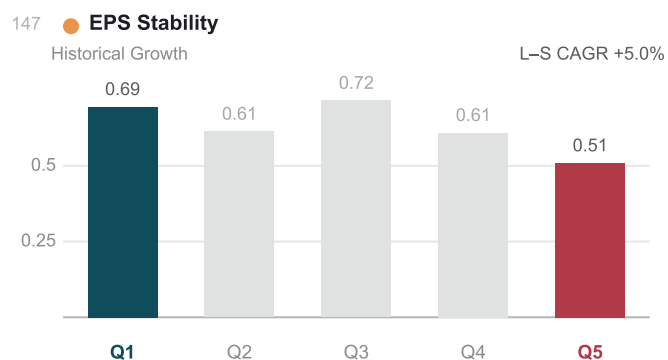
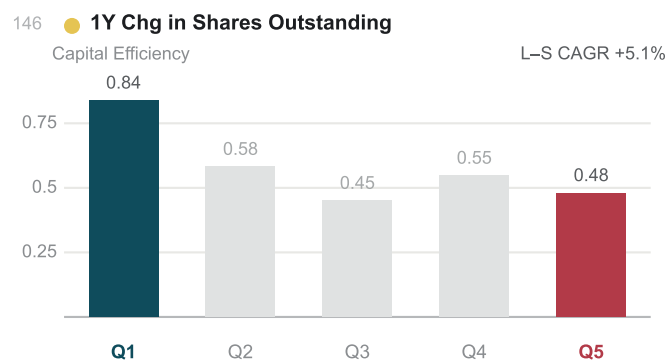
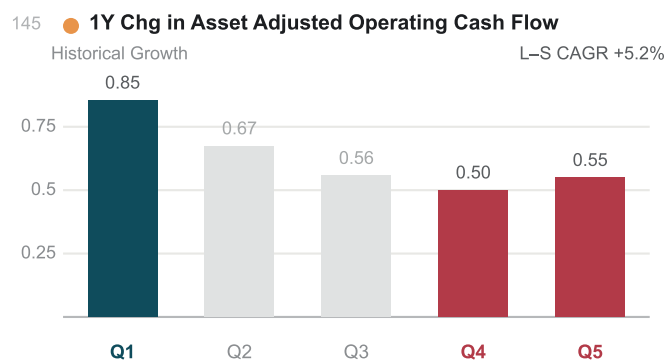


## 143 ● 12M - 1M Price Momentum

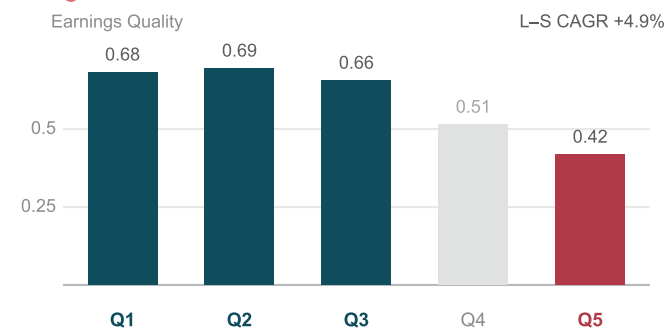


## 144 ● 1Y Chg in Free Cash Flow per Share

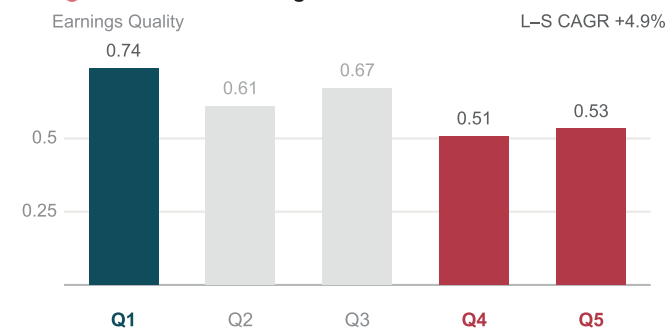




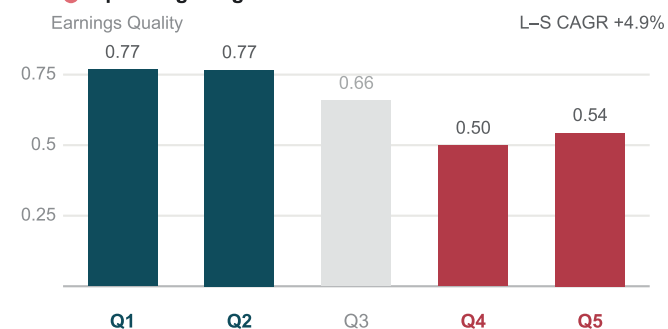
## 151 ● Free Cash Flow to Sales



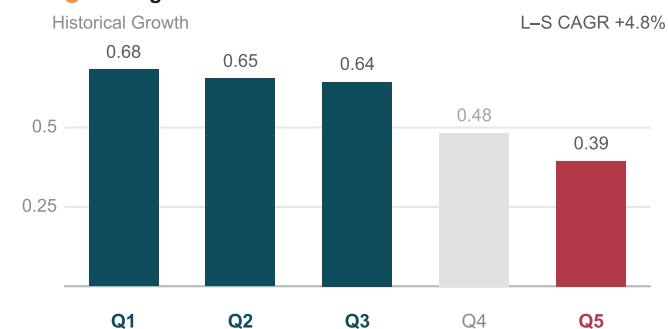
## 152 ● Year over Year Change of EPS to Sales



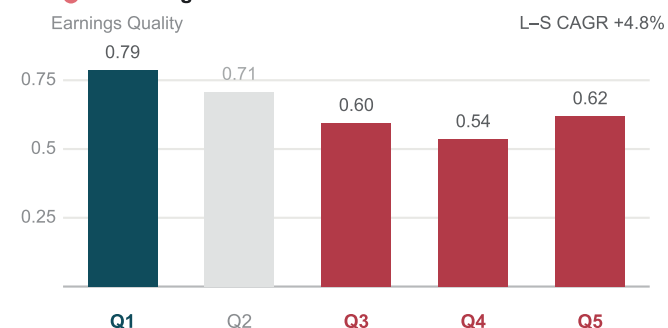
## 153 ● Operating Margin



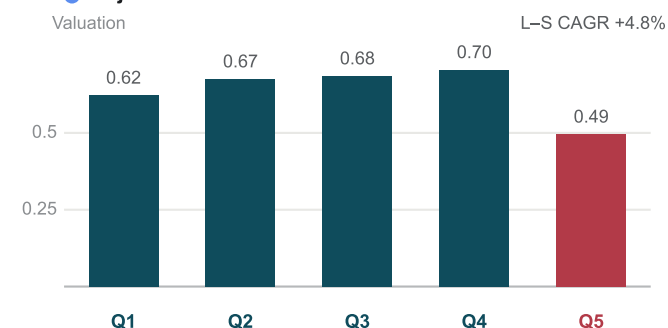
## 154 ● 3Y Chg in Sales to Price



## 155 ● EBIT Margin



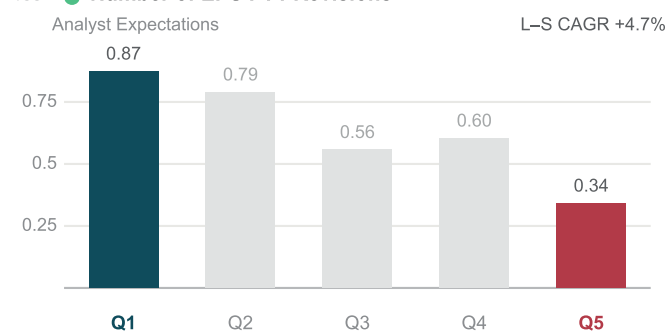
## 156 ● Adj Forward Free Cash Flow to Price Ratio



## 157 ● 3Y Chg in Free Cash Flow per Share



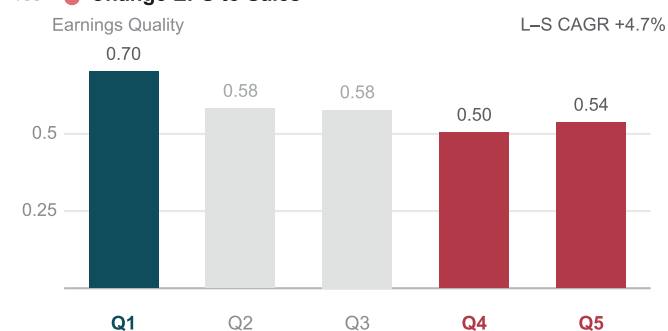
## 158 ● Number of EPS FY1 Revisions



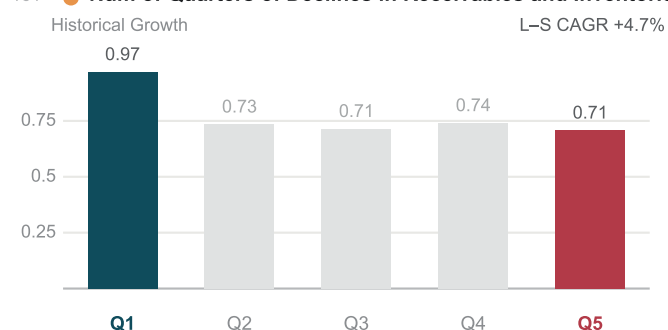
## 159 ● Working Capital to Assets Ratio



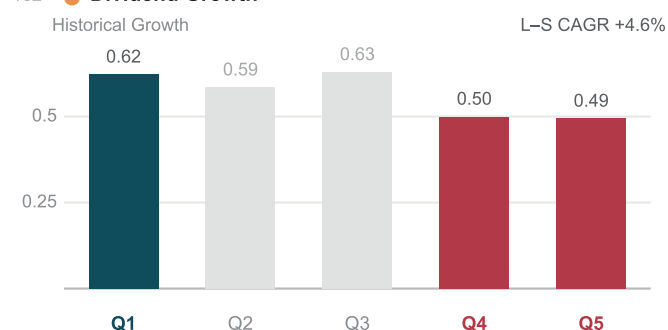
## 160 ● Change EPS to Sales

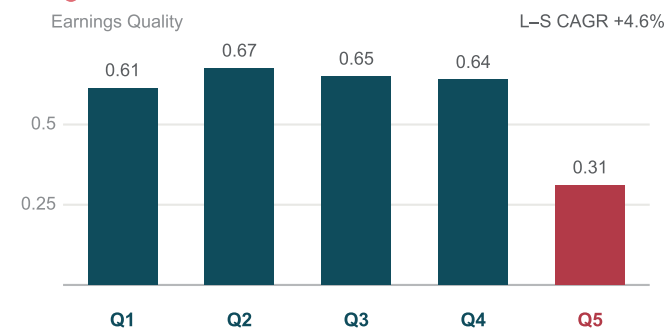
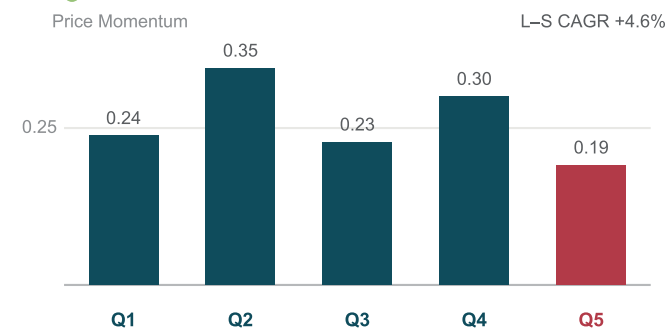
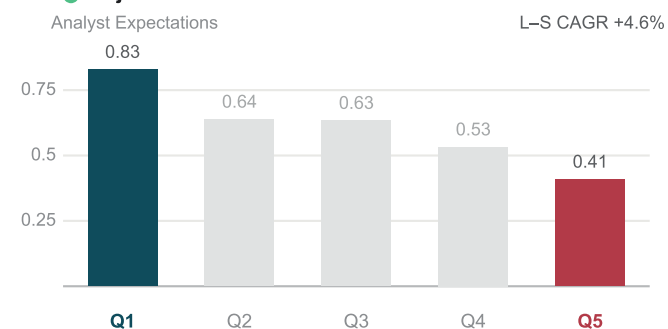
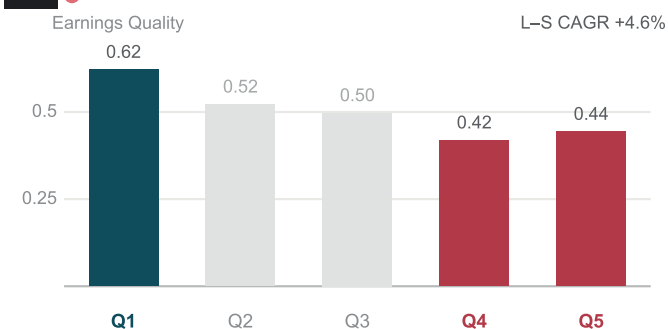
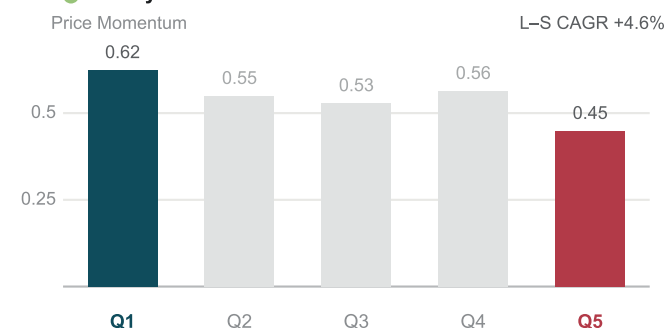
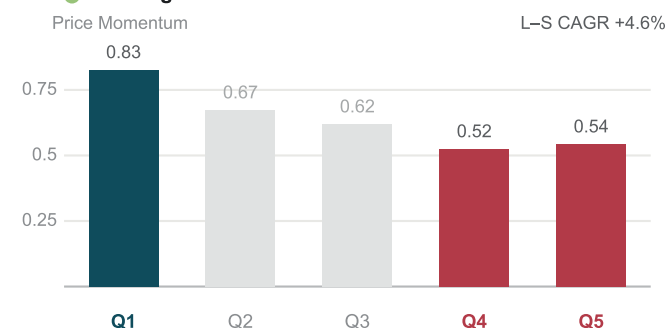


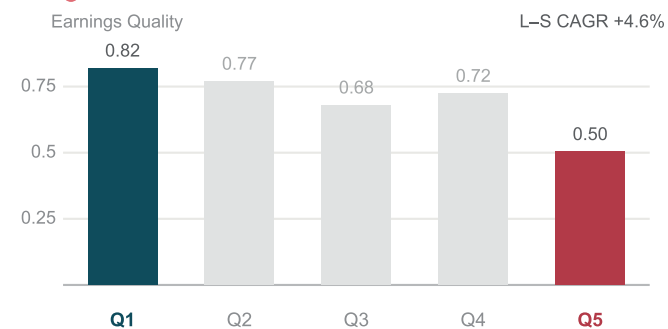
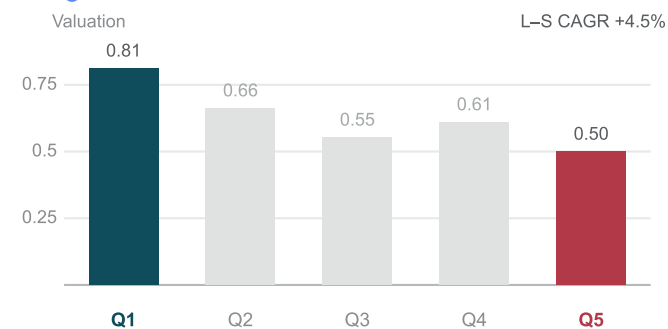
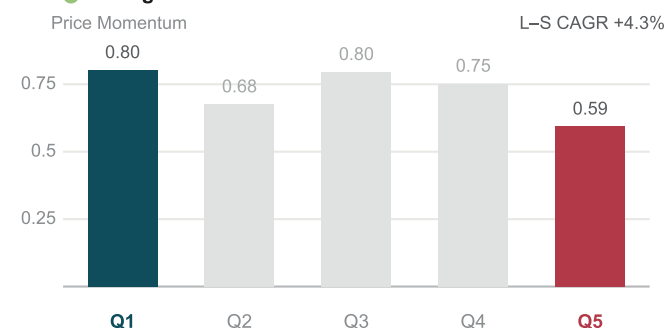
## 161 ● Num of Quarters of Declines in Receivables and Inventories/...

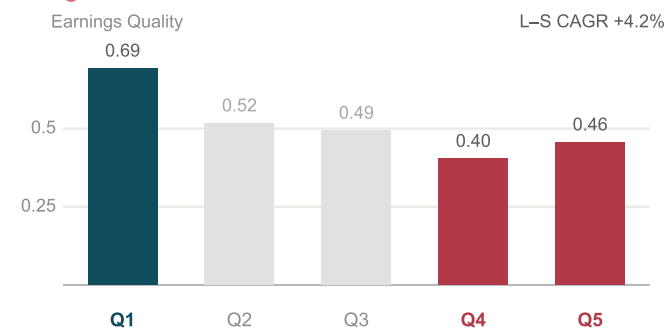
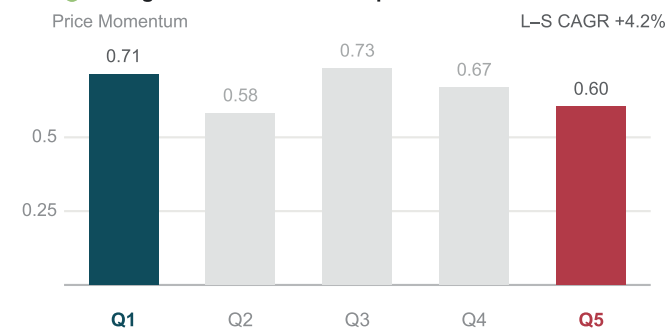
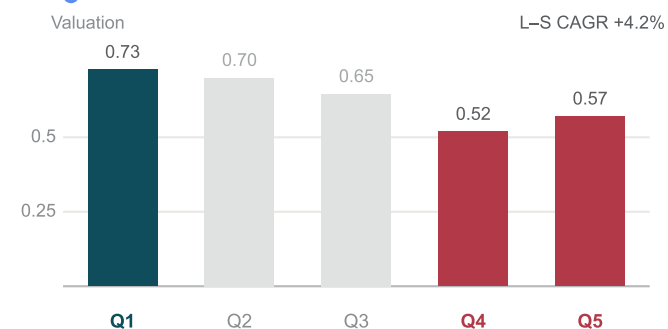
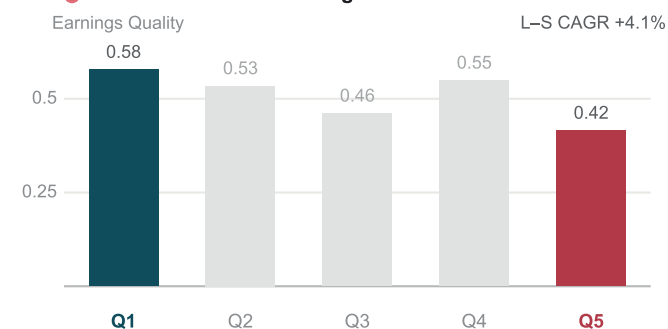
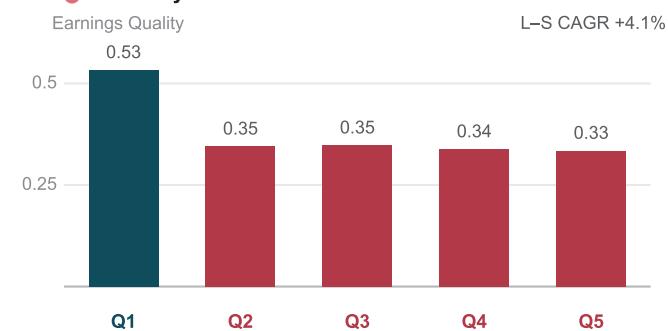


## 162 ● Dividend Growth



163 ● **Asset Turnover**164 ● **Share Turnover**165 ● **Adj Number of EPS FY2 Revisions**166 ● **Accrual Ratio - Cash Flows**167 ● **60 Day Coefficient of Variation of Volume to Price**168 ● **52W High Low**

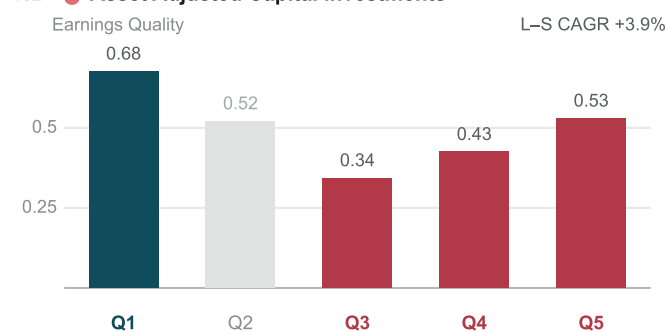
169 ● **Current Ratio**170 ● **Cash to Total Assets**171 ● **3Y Chg in Sales Turnover**172 ● **6M Momentum in Trailing 12M Sales**173 ● **Change in Number of Holders**174 ● **3Y Chg in Asset Adjusted Free Cash Flow**

175 ● **SG&A to Sales**176 ● **Foreign Institution Ownership**177 ● **Net Current Assets to Price Ratio**178 ● **Sales to Gross Profit Margin Growth**179 ● **Risk Adjusted Relative Strength**180 ● **Inventory to Assets Ratio**

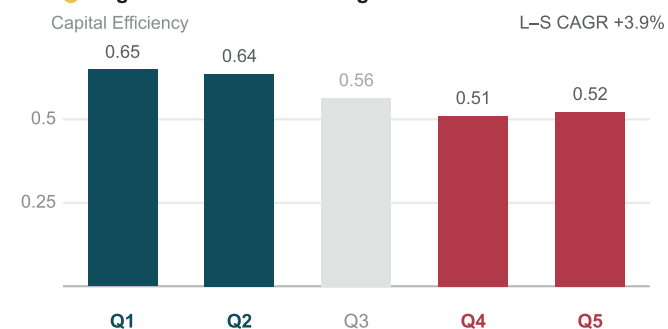
## 181 ● ROA 60 Month Slope



## 182 ● Asset Adjusted Capital Investments



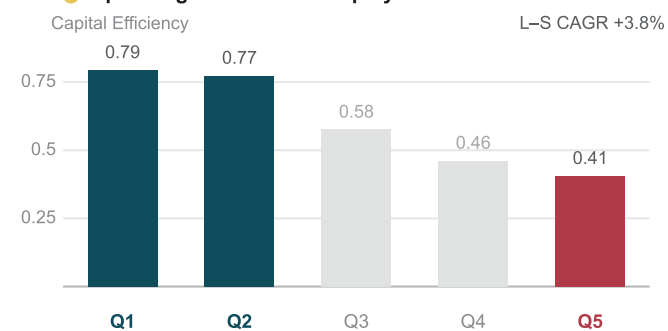
## 183 ● Degree of Financial Leverage



## 184 ● Return on Equity Stability

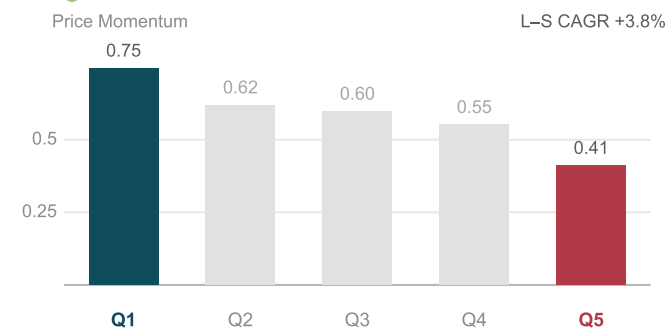
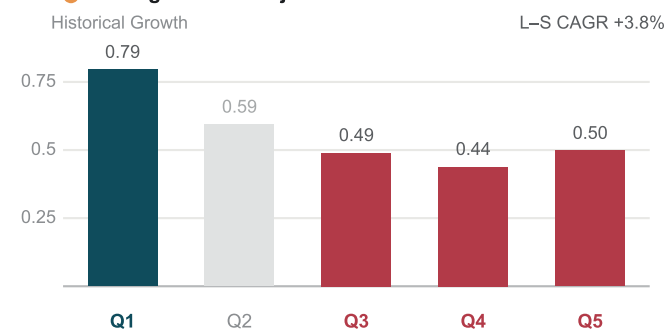
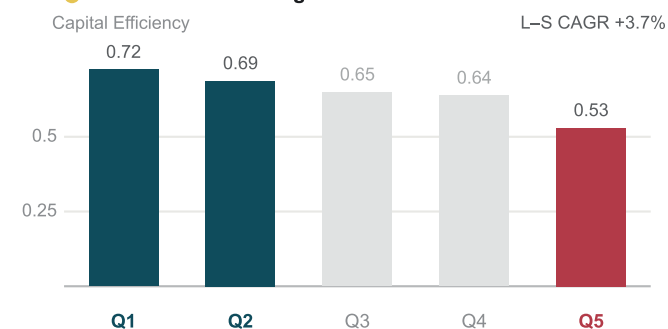
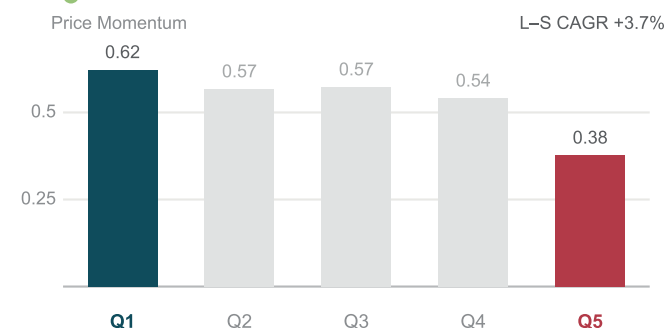
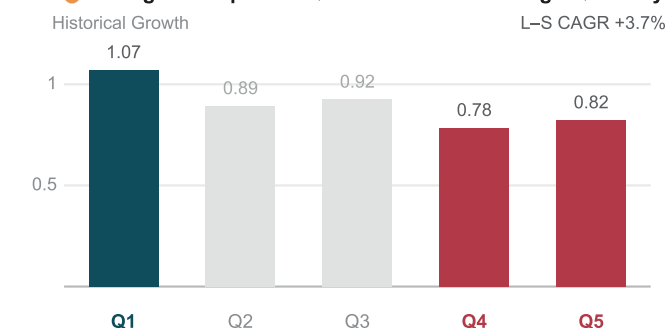


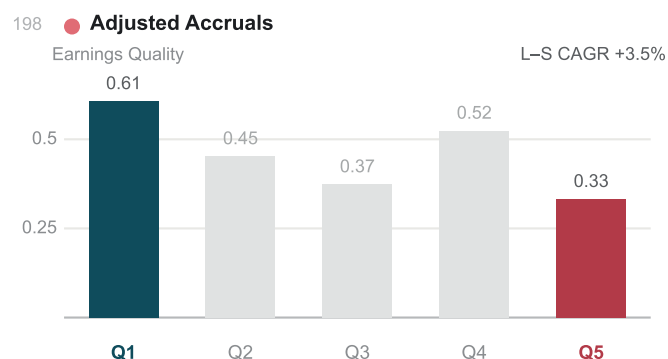
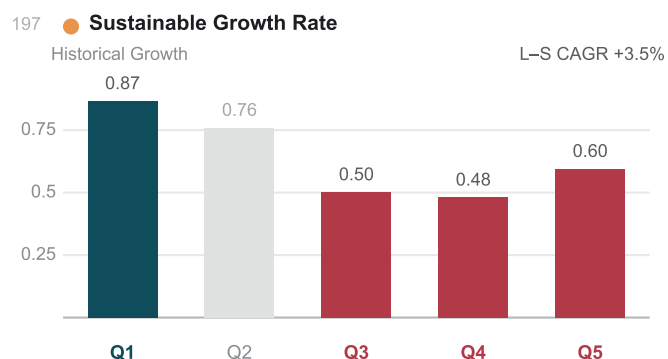
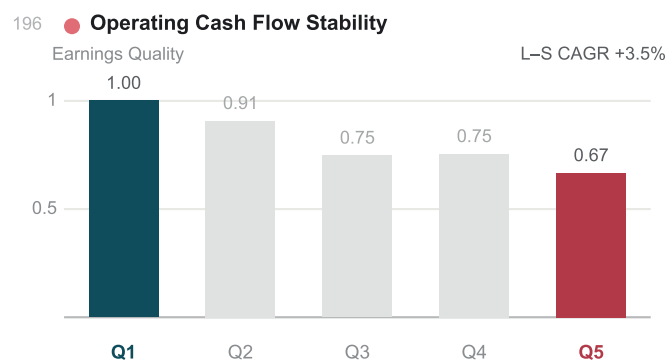
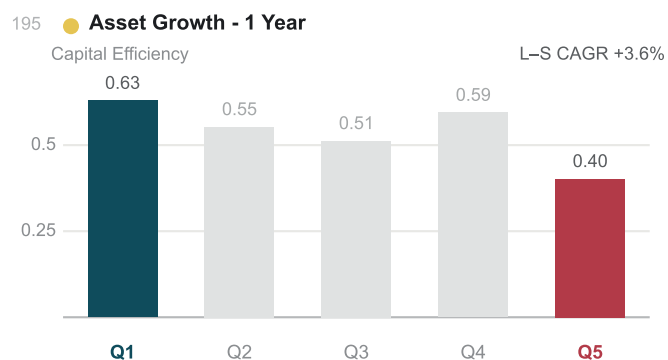
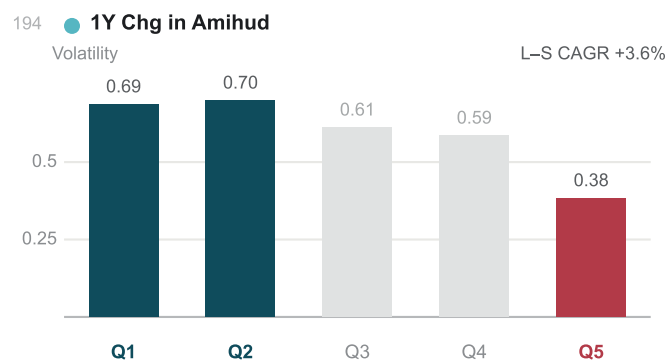
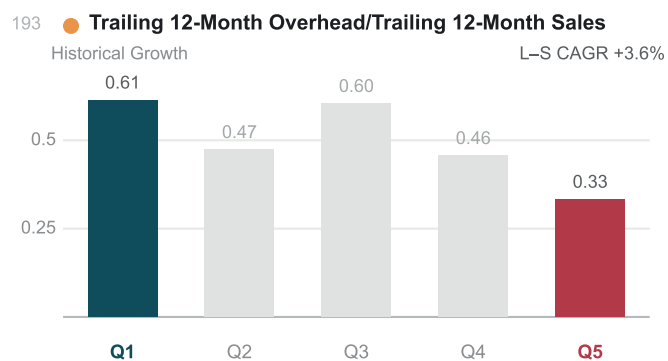
## 185 ● Operating Cash Flow to Equity

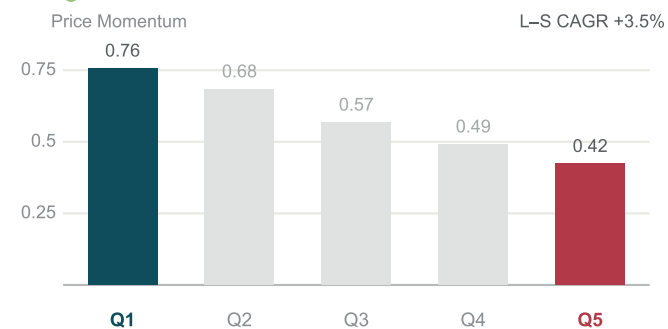
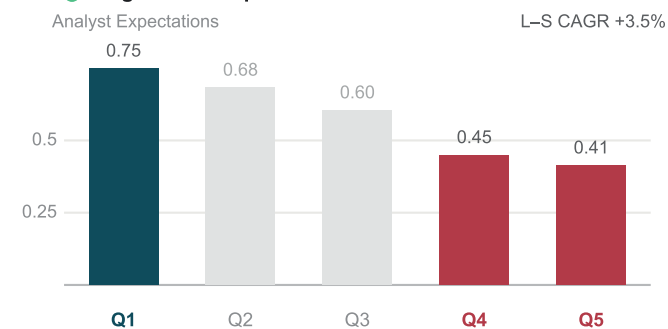
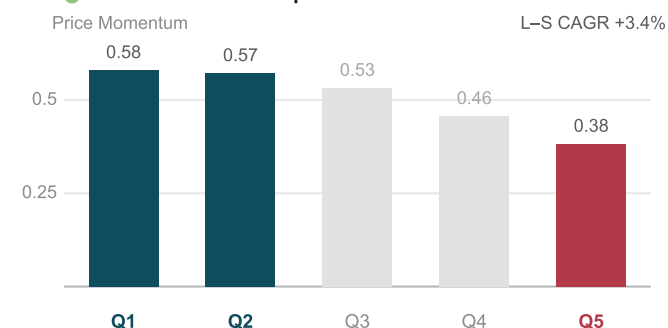


## 186 ● Retained Earnings to Total Assets



187 ● **Net Profit Margin**188 ● **26W Price Stat**189 ● **3Y Chg in Asset Adjusted EPS**190 ● **Year over Year Change of Total Debt**191 ● **24M Residual Return Variance**192 ● **Change in Slope of 4-Quarter Trendline through Quarterly Sales**



199 ● **180D Price TStat**200 ● **Target Price Gap to 6M EMA**201 ● **Free Cash Flow to Common Equity**202 ● **1Y Chg in Earnings per Share**203 ● **Working Capital to Total Assets**204 ● **Institution Ownership Concentration**

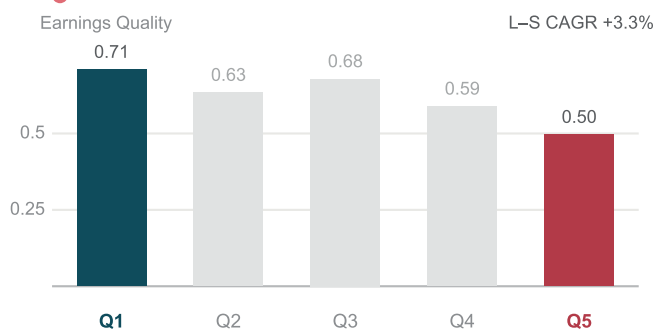
## 205 ● Return on Assets Stability



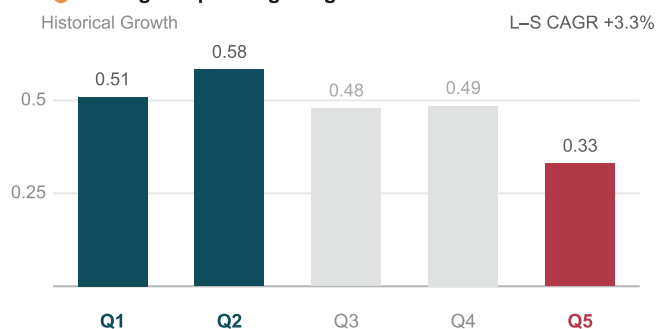
## 206 ● 3Y Chg in Earnings per Share



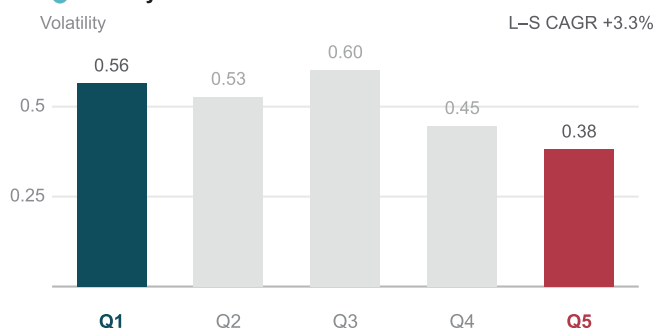
## 207 ● Accrual Ratio - Balance Sheet



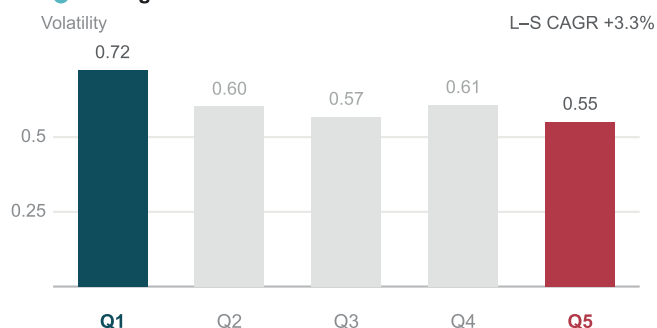
## 208 ● 3Y Chg in Operating Margin



## 209 ● 130 Day Minimum Return

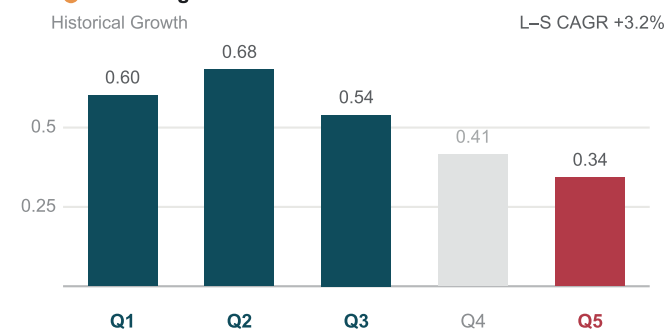
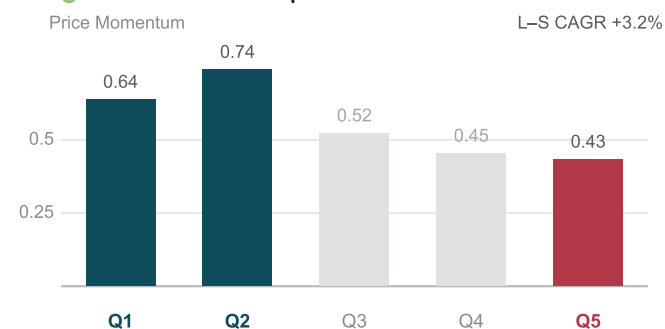
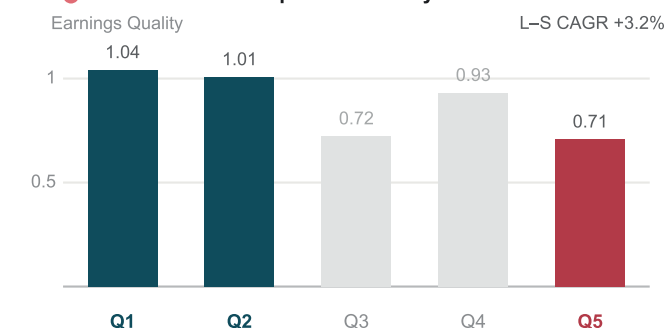
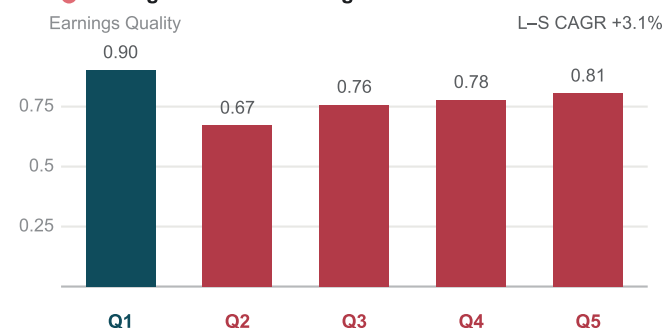


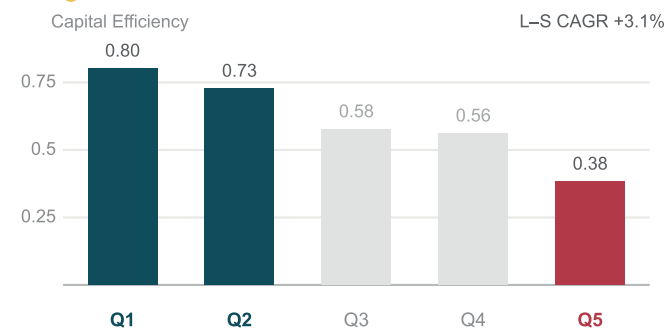
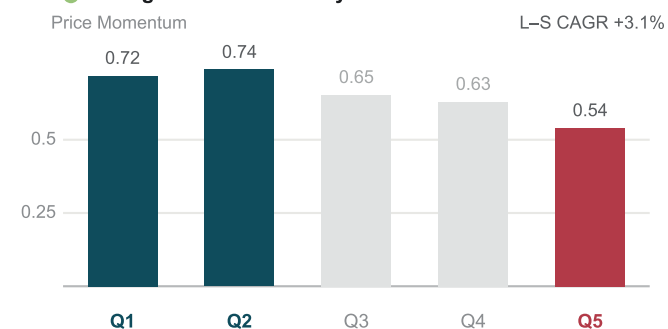
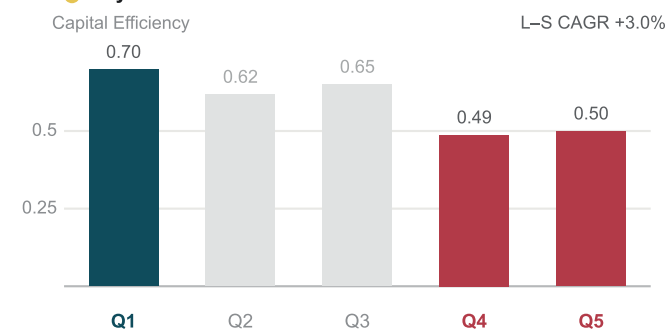
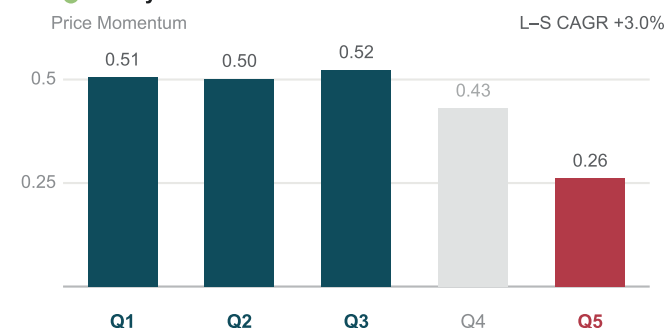
## 210 ● 1Y Chg in Debt to Assets



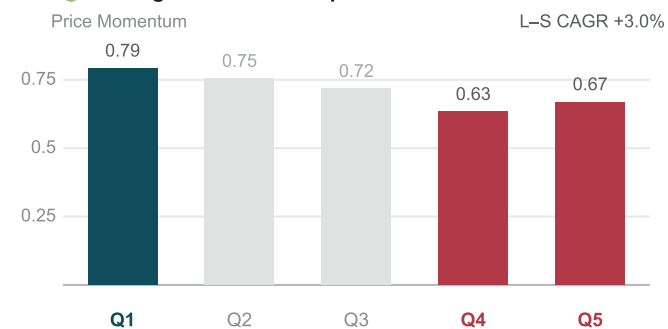
**211 Ownership Breadth Stability of Institution Ownership by Buyers**

**212 Pretax Profit Margin**

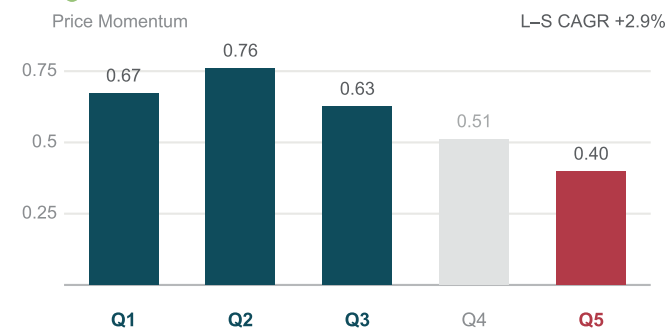
**213 3Y Average Annual Sales Growth**

**214 Institution Ownership Level**

**215 Research & Development Intensity**

**216 1Y Chg in Sales to Earnings**


217 ● **Gross Profit to Assets**218 ● **Cash Conversion Cycle**219 ● **Change in Number of Buyers**220 ● **Payout Ratio**221 ● **10 Day MACD Trend**222 ● **Change of Sales to Change of EPS**

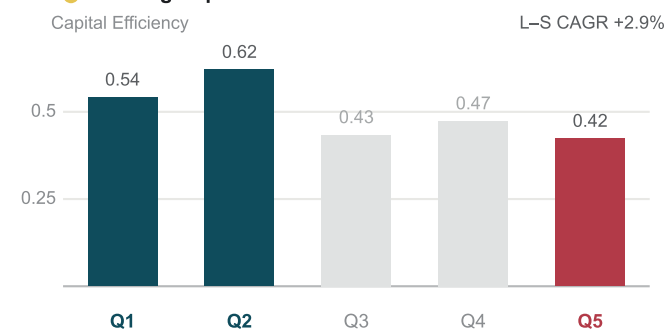
## 223 ● 6M Chg in 18M CAPM Alpha



## 224 ● Investment Duration



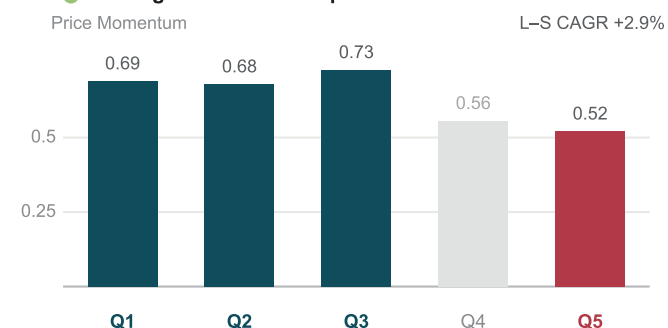
## 225 ● Working Capital Turnover Ratio



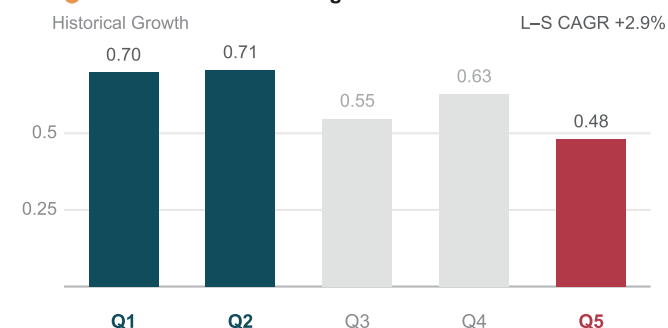
## 226 ● Ownership Breadth Stability of Institution Ownership by Hol...



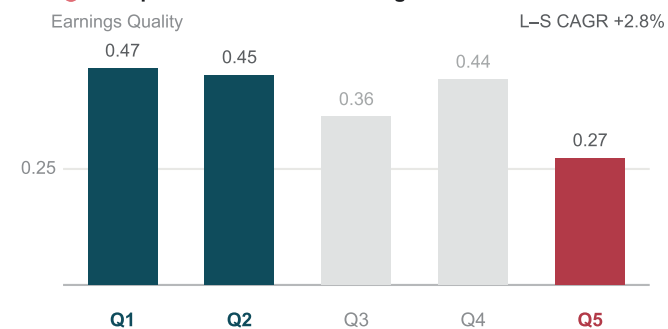
## 227 ● 6M Chg in 36M CAPM Alpha



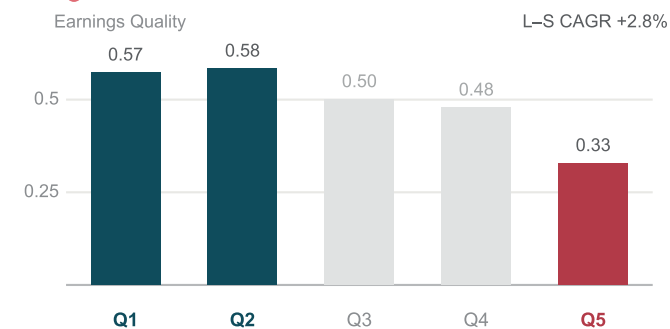
## 228 ● 3M Momentum in Trailing 12M Sales



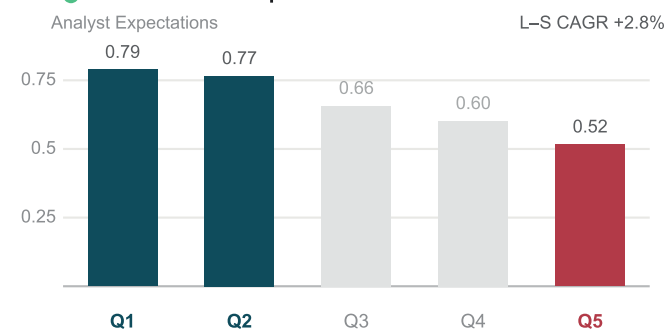
## 229 ● Unexpected Receivables Change



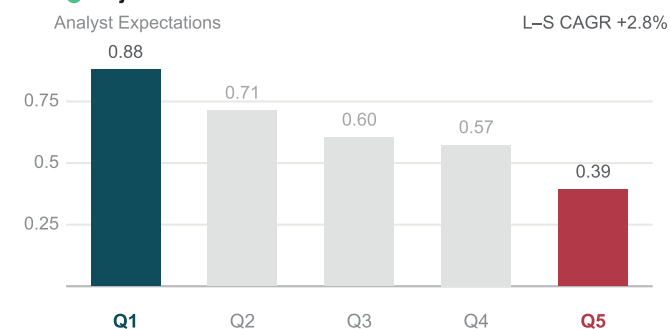
## 230 ● Cash to Sales



## 231 ● EBIT Estimate Dispersion



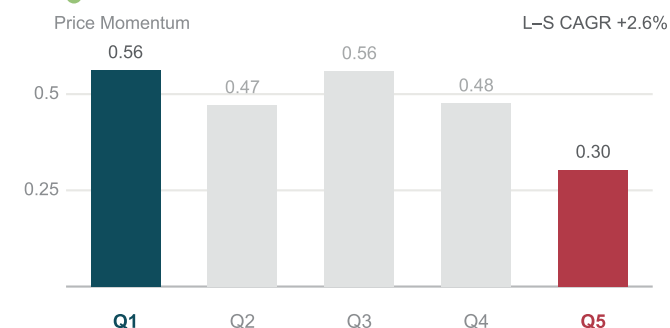
## 232 ● Adjusted Number of EPS FY1 Revisions



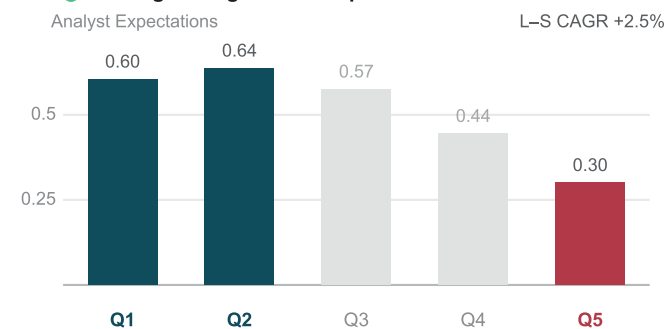
## 233 ● Volatility adjusted 12M return



## 234 ● Standardized 36M Residual



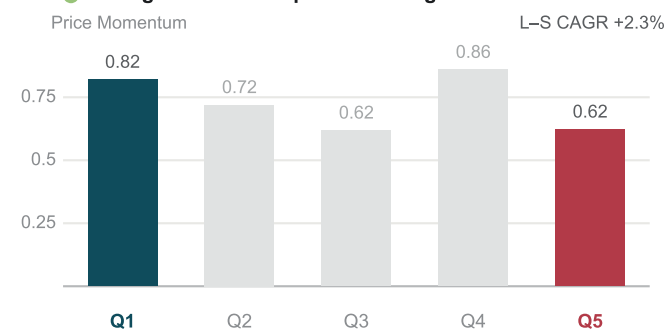
## 235 ● 6M Chg in Target Price Gap



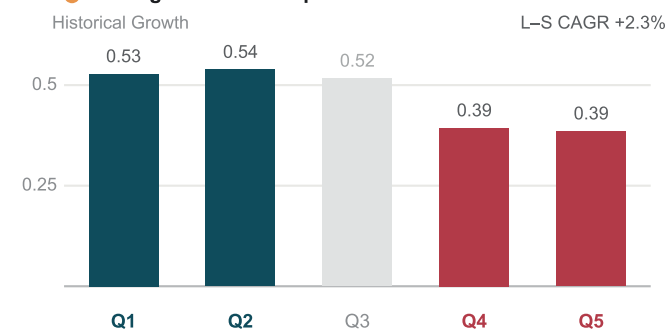
## 236 ● 1Y Chg in EPS to Operating Cash Flow



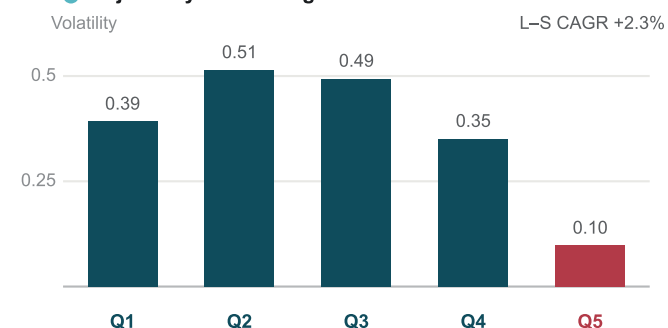
## 237 ● Change of Ownership Level - Hedge Fund



## 238 ● 1Y Chg in Cash Flow per Share

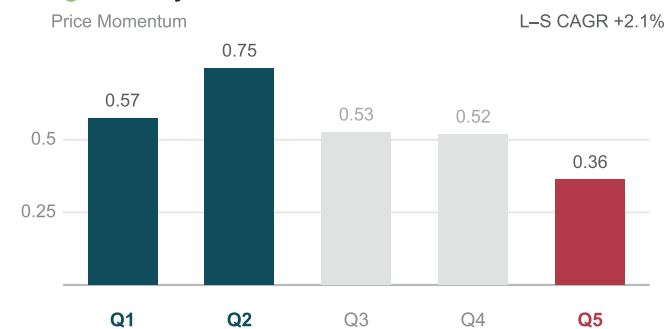
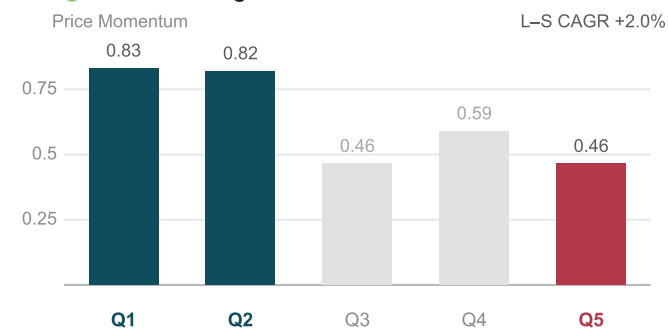
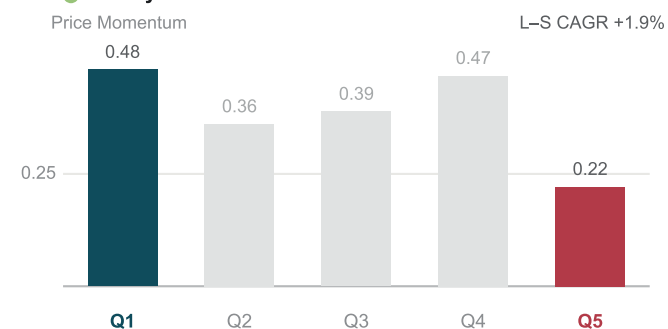
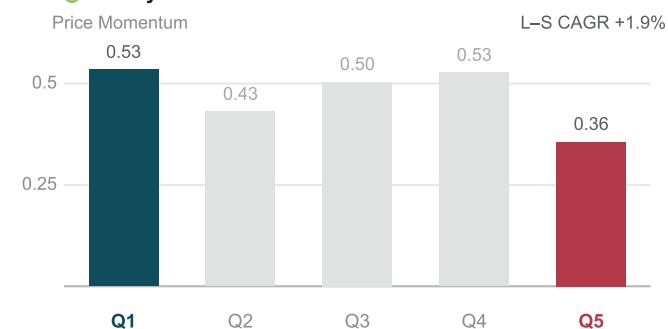
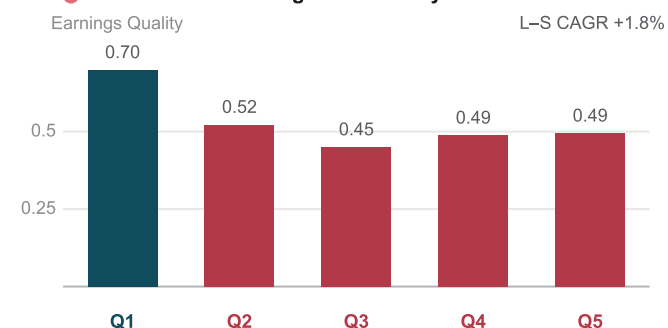
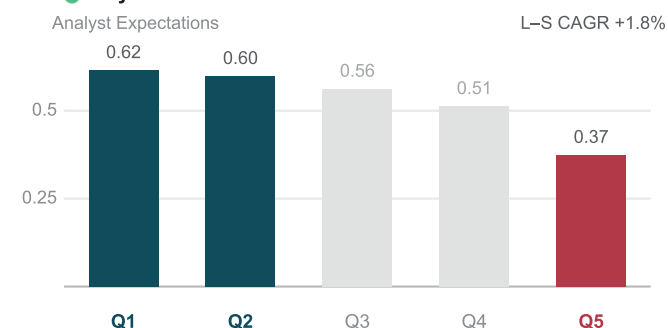


## 239 ● Adj 50 Day Volume Signal



## 240 ● Analyst Earnings Estimate Diffusion

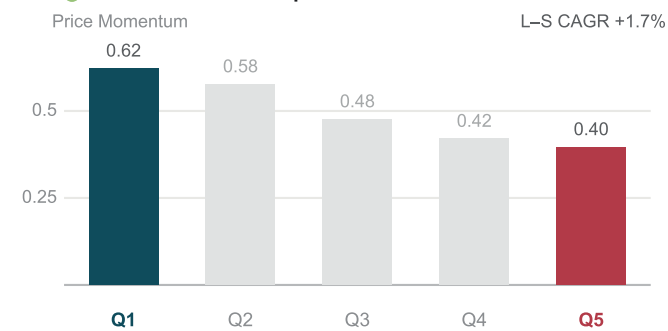


241 ● **Max Daily Return in the Past 6 Months**242 ● **6 Month Average Share Turn Over**243 ● **30 Day Coefficient of Variation of Volume to Price**244 ● **20 Day Coefficient of Variation of Volume to Price**245 ● **Year over Year Change of Inventory to Assets**246 ● **Buy-to-Sell Recommendation Ratio Less 3M SMA**

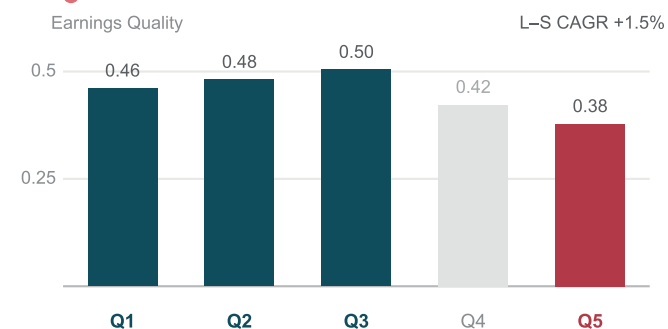
## 247 ● Closing Price to 52 Week High



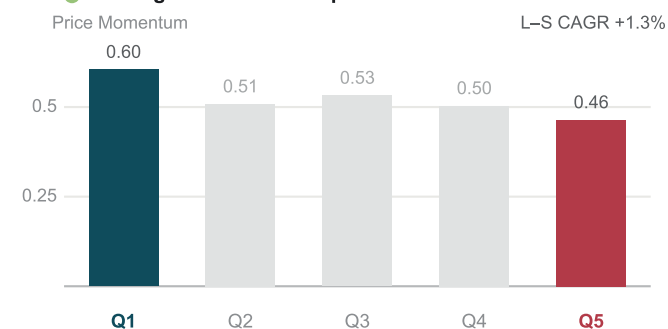
## 248 ● Institution Ownership Turnover



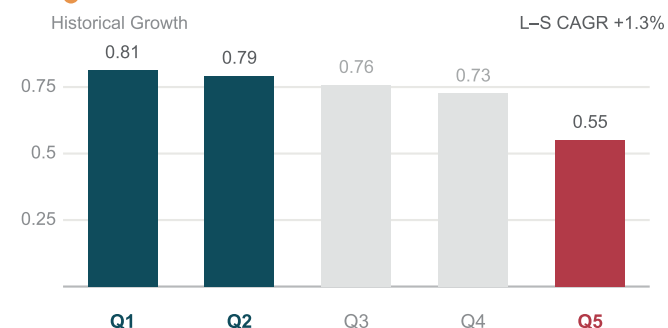
## 249 ● ROE 20 Qtr Standard Deviation



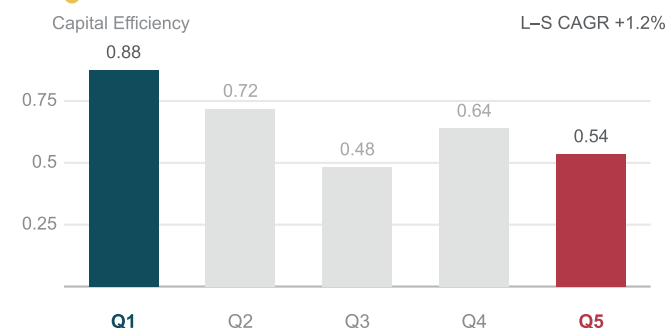
## 250 ● 6M Chg in 12M CAPM Alpha



## 251 ● Sales Acceleration



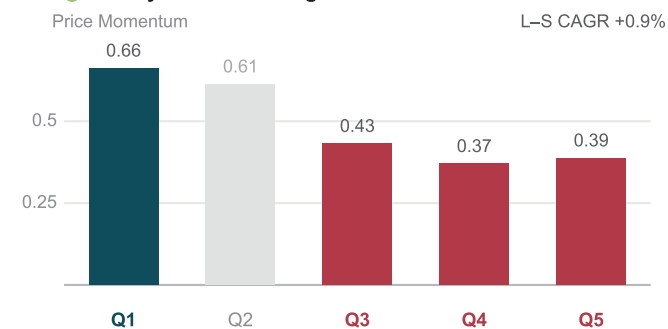
## 252 ● Normalized ROA



## 253 ● 6M Avg Chg 1M Recommendation



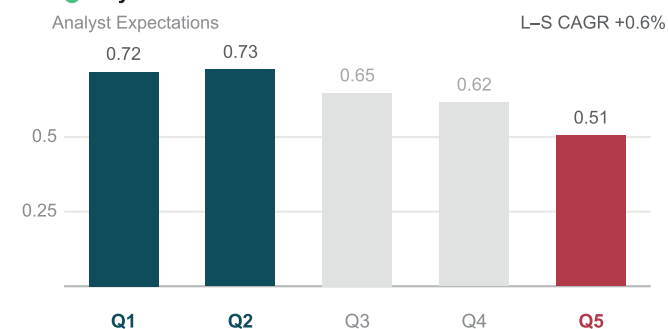
## 254 ● 14-Day Relative Strength Index



## 255 ● Change in TTM Depr. to Capex



## 256 ● Buy-to-Sell Recommendation Ratio Less 3M EMA

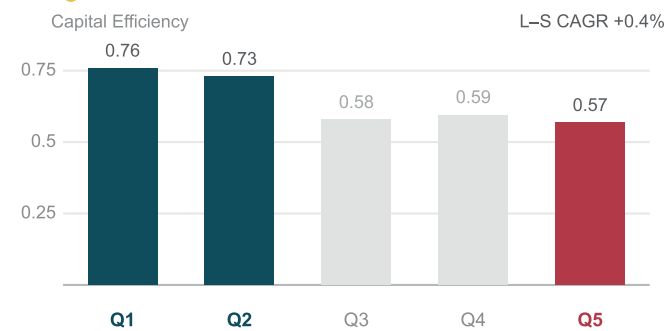
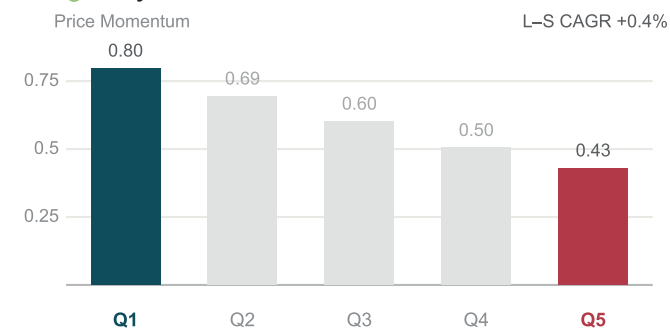
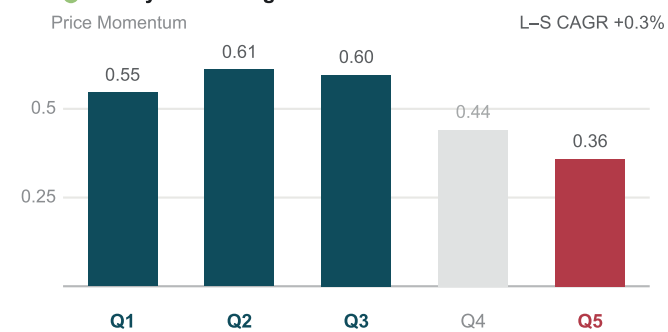
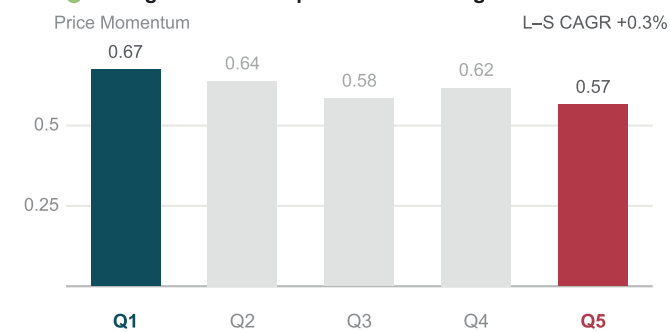
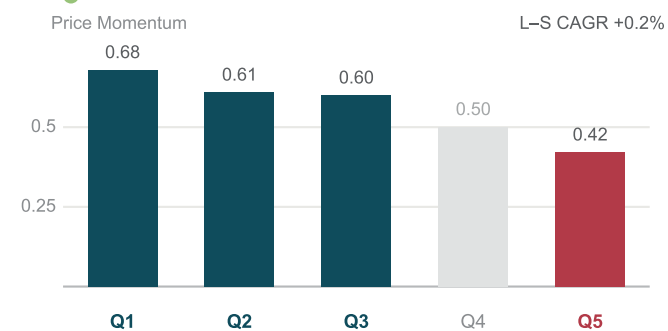


## 257 ● Revenue Estimate Dispersion



## 258 ● 8-Week Change in 12-Month Forward Earnings Consensus Estima...



259 ● **Normalized ROE**260 ● **5 Day Price Reversal**261 ● **50 Day Volume Signal**262 ● **Change of Ownership Level - All Managers**263 ● **1M Price Reversal**264 ● **1Y Chg in Share Turnover**